



SOAL UKK TKJ PAKET 4 2024

MID SEMESTER MPP

— KELOMPOK ? —



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Dirly Marentek

LOGIN KE WINBOX KEMUDIAN CONNECT

WinBox v3.32 (TEST)

File Tools

Connect To:

Login:

Password:

☒ Keep Password

☐ Open In New Window

☒ Auto Reconnect

Managed Neighbors

MAC Address	IP Address	Identity	Version	Board	Uptime
18 FD 74 8F 51 A6	192.168.1.6	RouterOS	6.48.6 (p)	RB941-2nD	00:07:26

RESET CONFIGURATION

Reset Configuration

☐ Keep User Configuration

☐ CAPS Mode

☒ No Default Configuration

☐ Do Not Backup

Run After Reset:

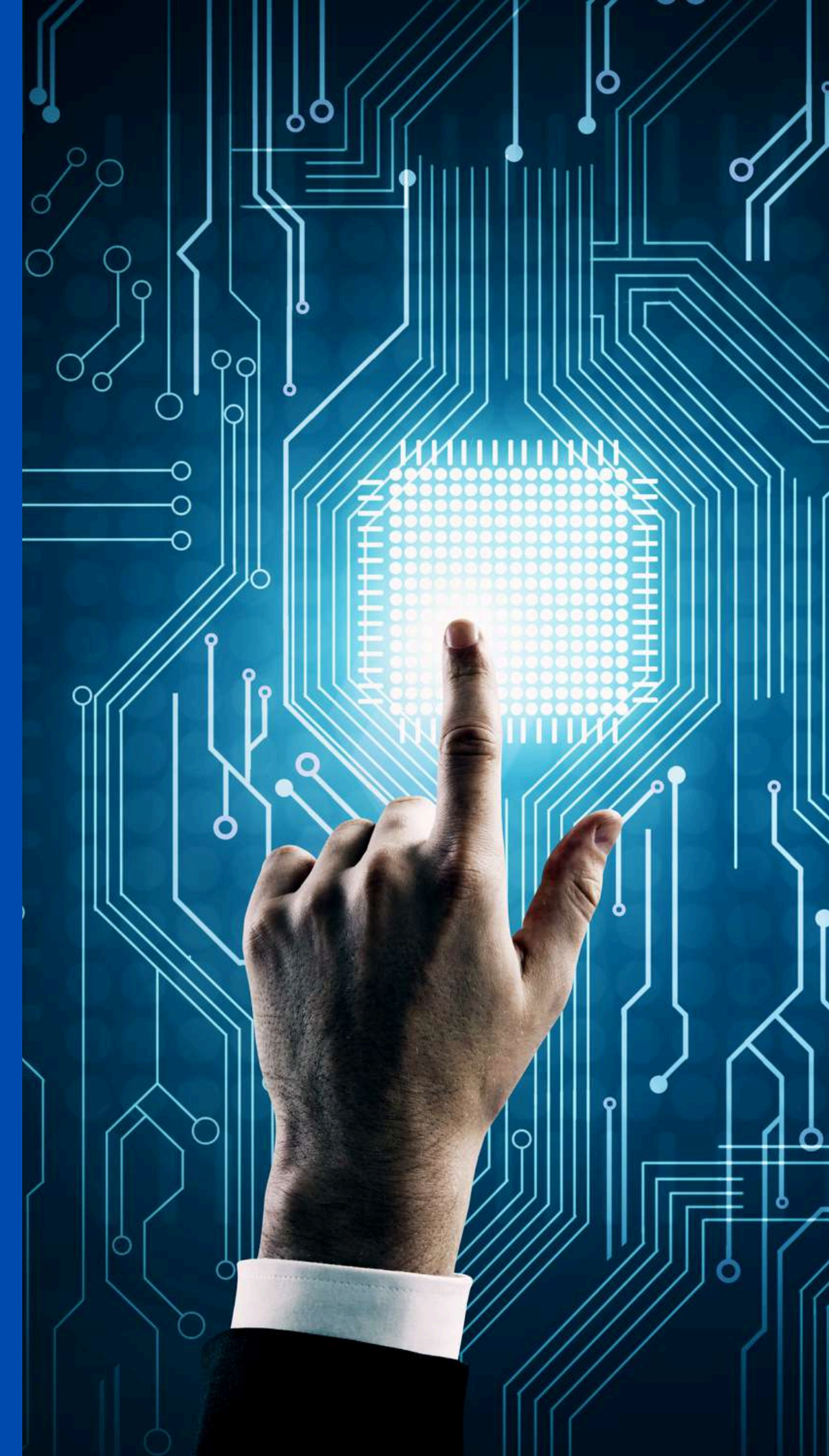
Reset Configuration

Cancel

Reset Configuration

Do you want to reset all your configuration and reboot?

Yes No



**LOGIN ULANG KE
WINBOX
KEMUDIAN KLIK
“CONNECT”**

WinBox (64bit) v3.40 (Addresses)

File Tools

Connect To: 18:FD:74:20:0C:A5

Login: admin

Password:

Add/Set

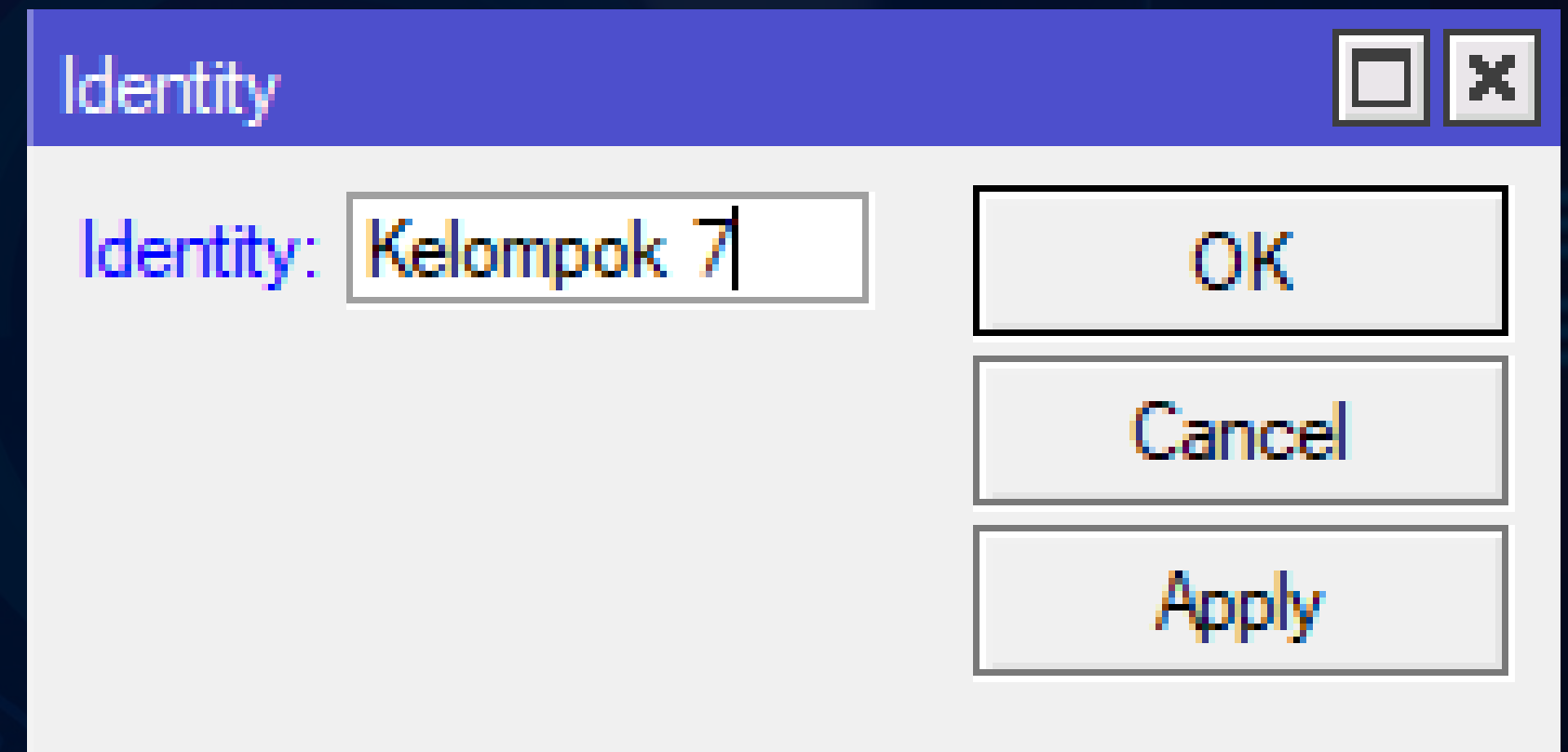
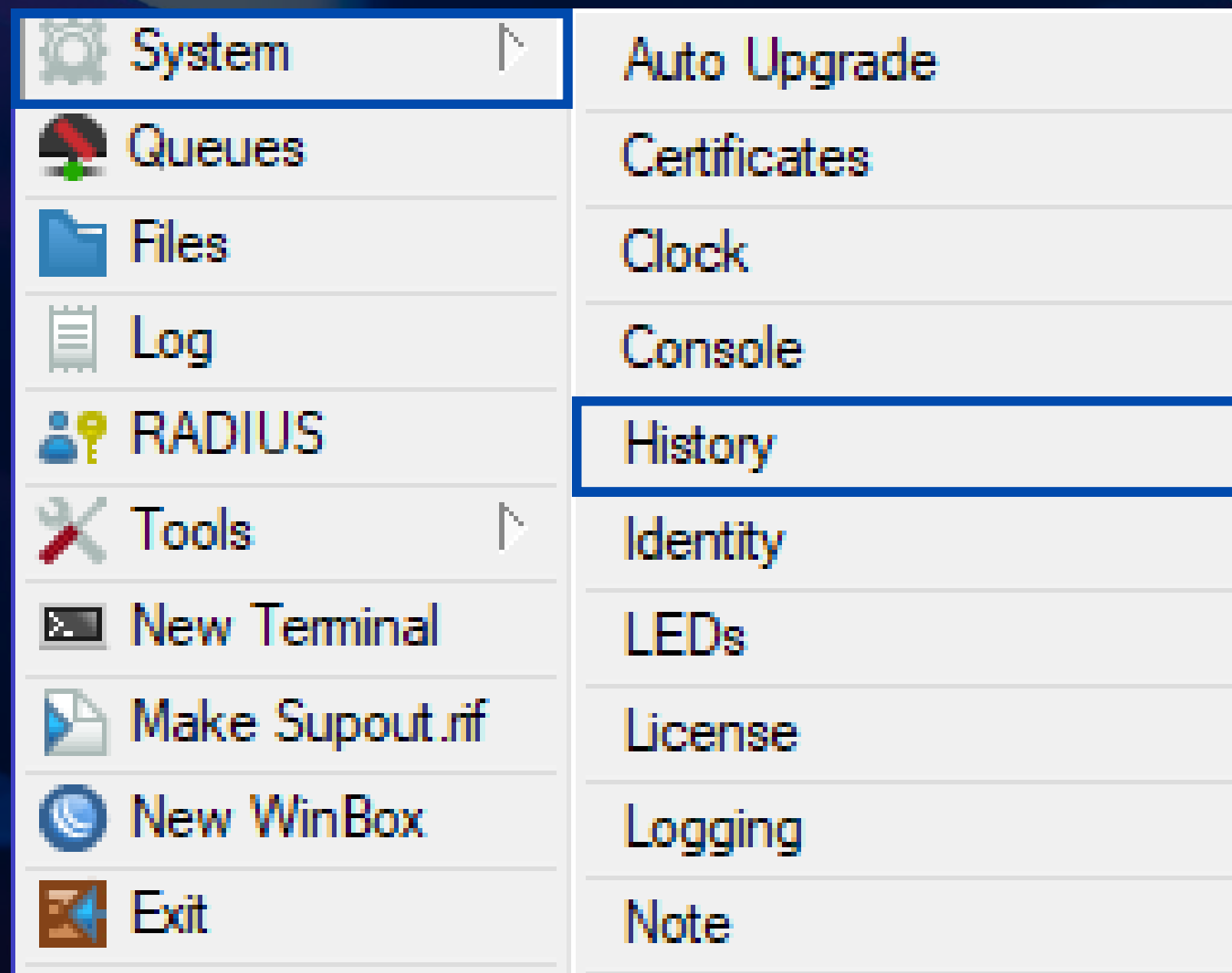
Managed Neighbors

Refresh

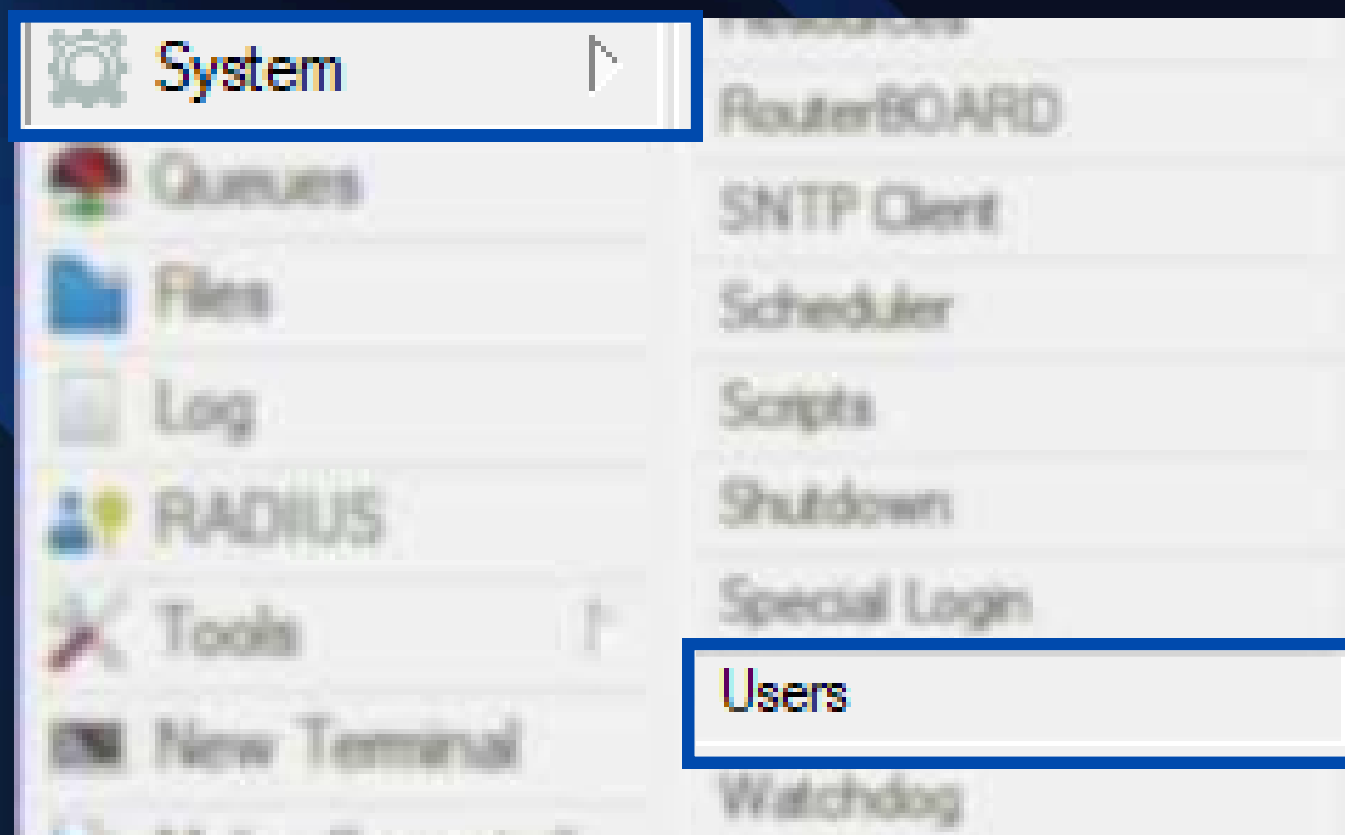
MAC Address in

MAC Address	IP Address	Identity	Version
18:FD:74:20:0C:A5	0.0.0.0	RouterOS	6.48.6 (lo

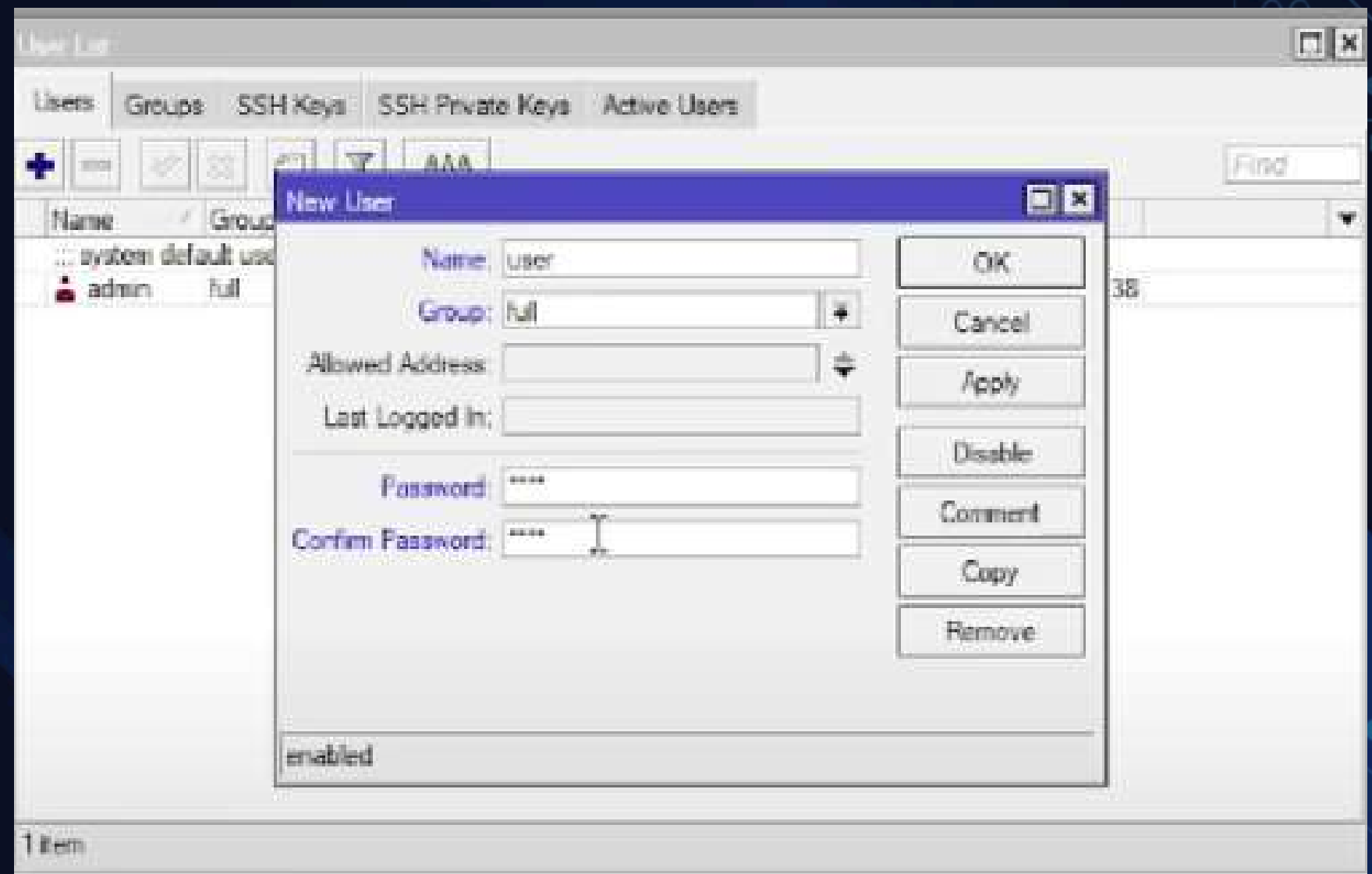
UBAH IDENTITY



TAMBAHKAN USER BARU

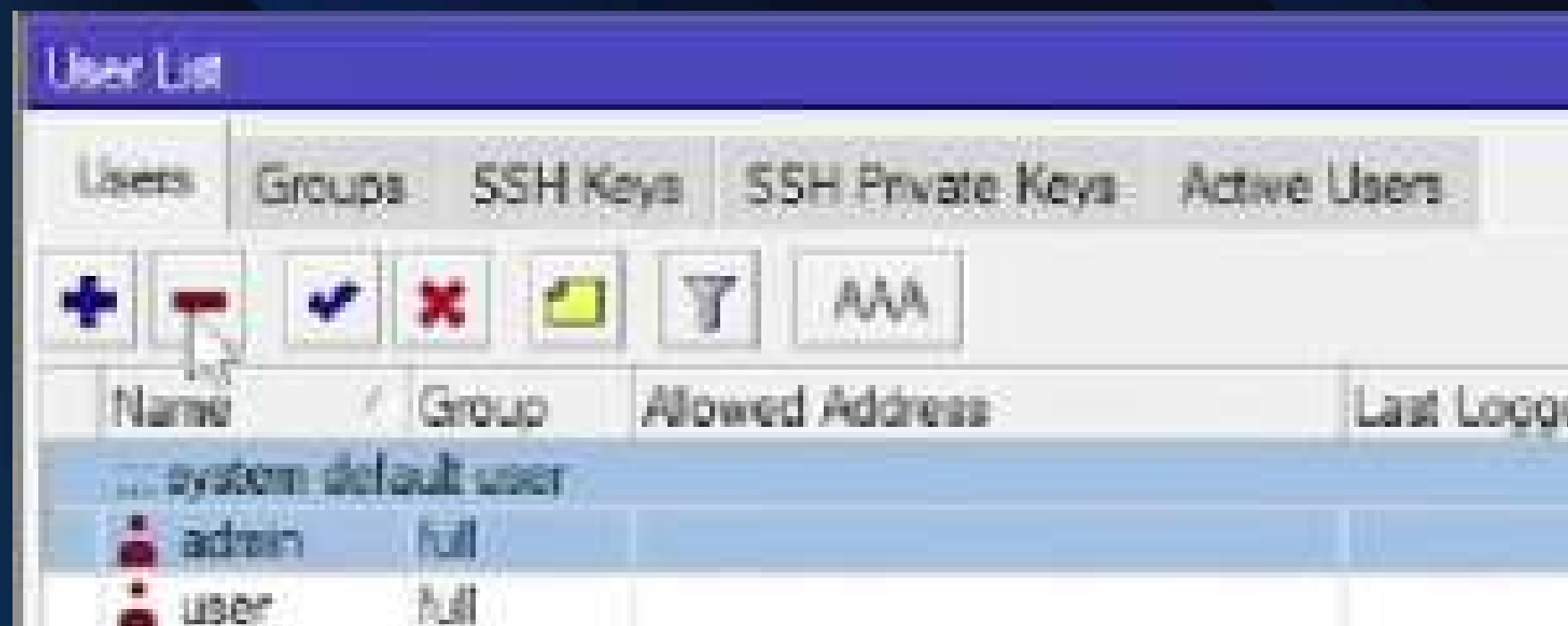
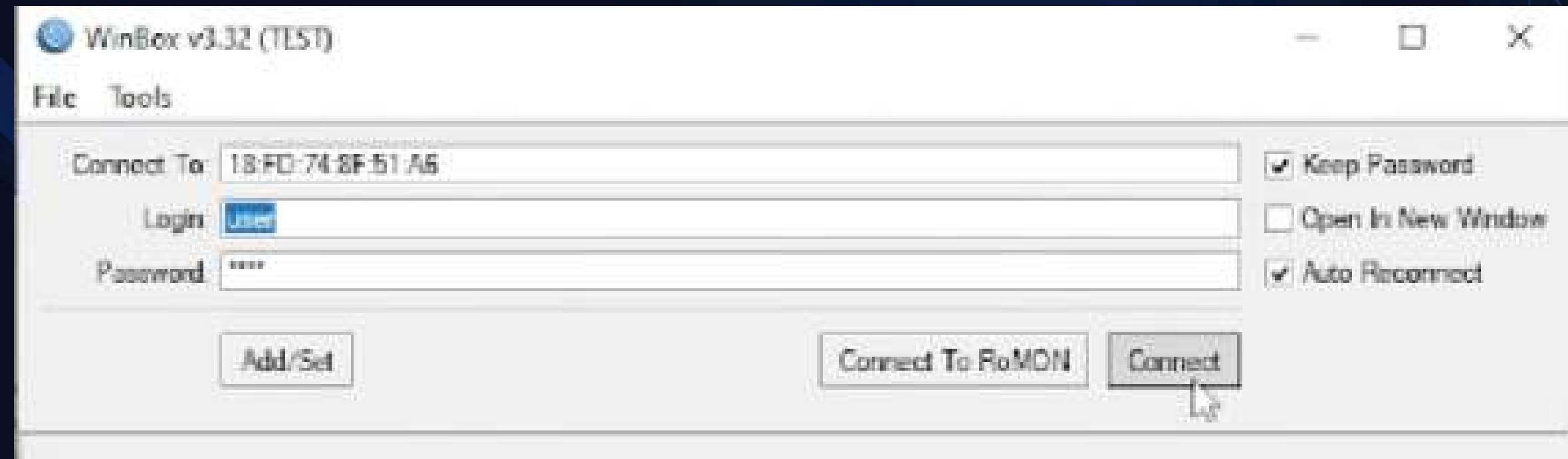
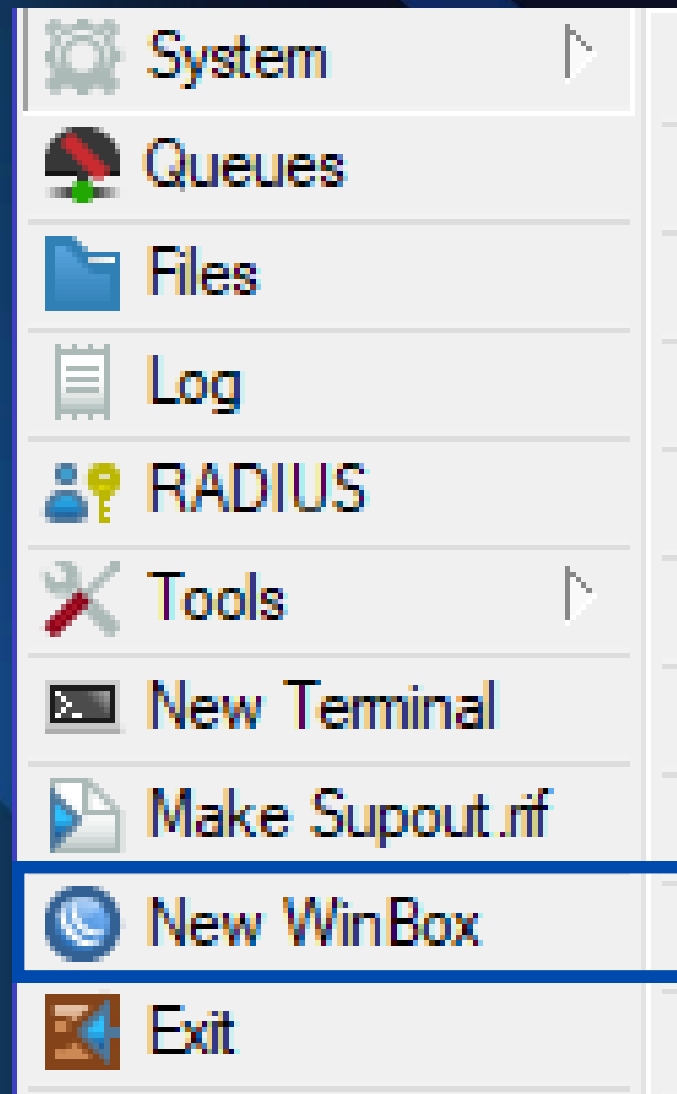


Name : user
Group : full
Password : 1234



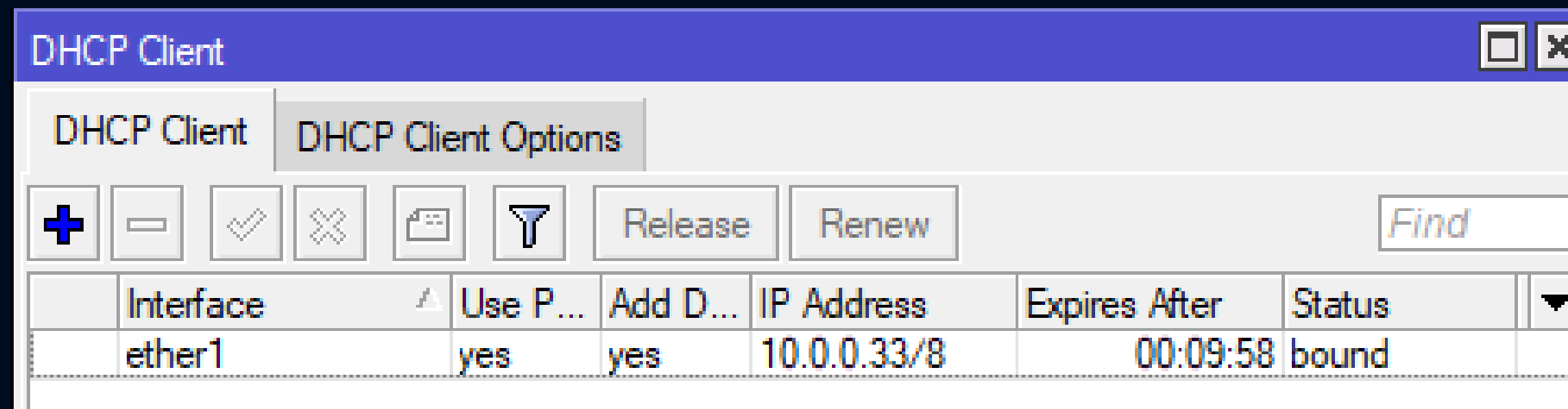
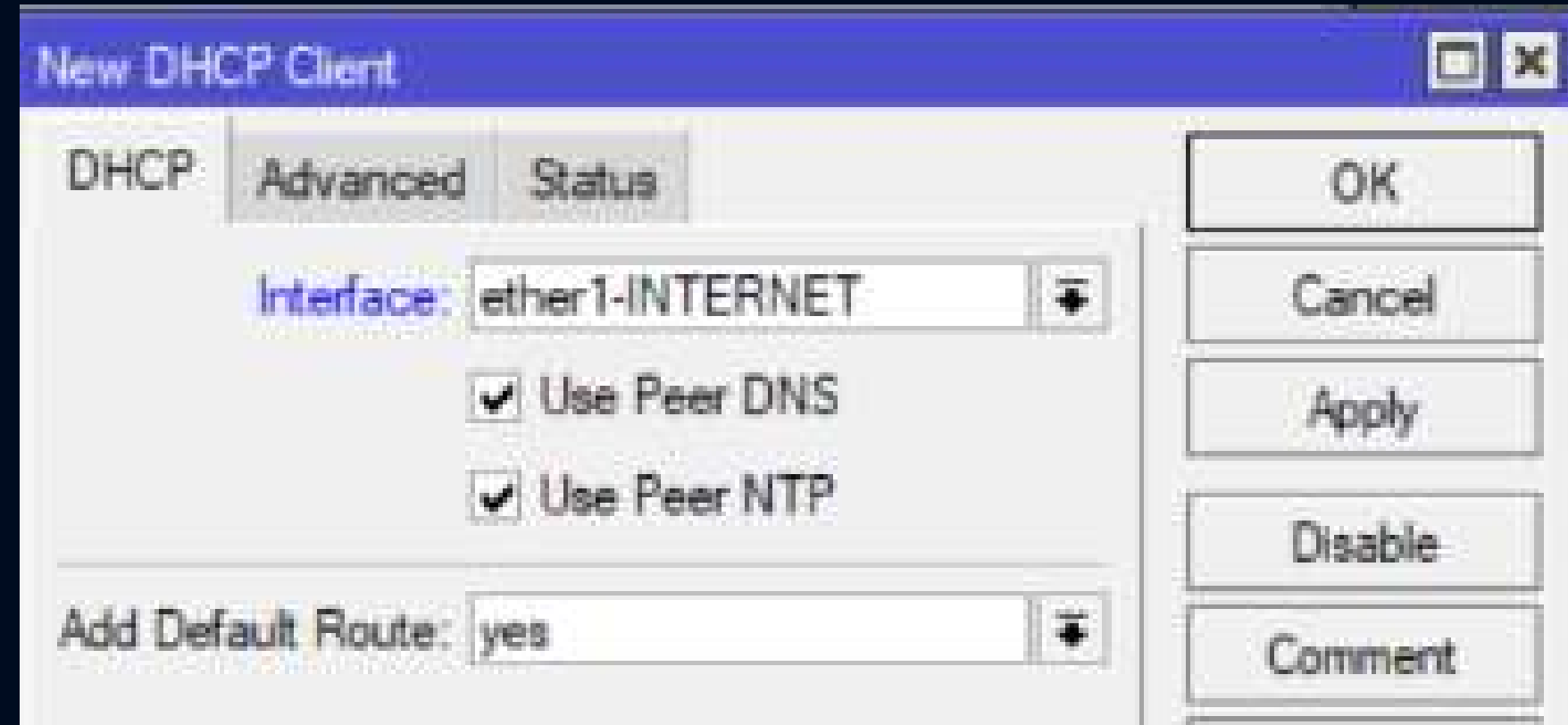
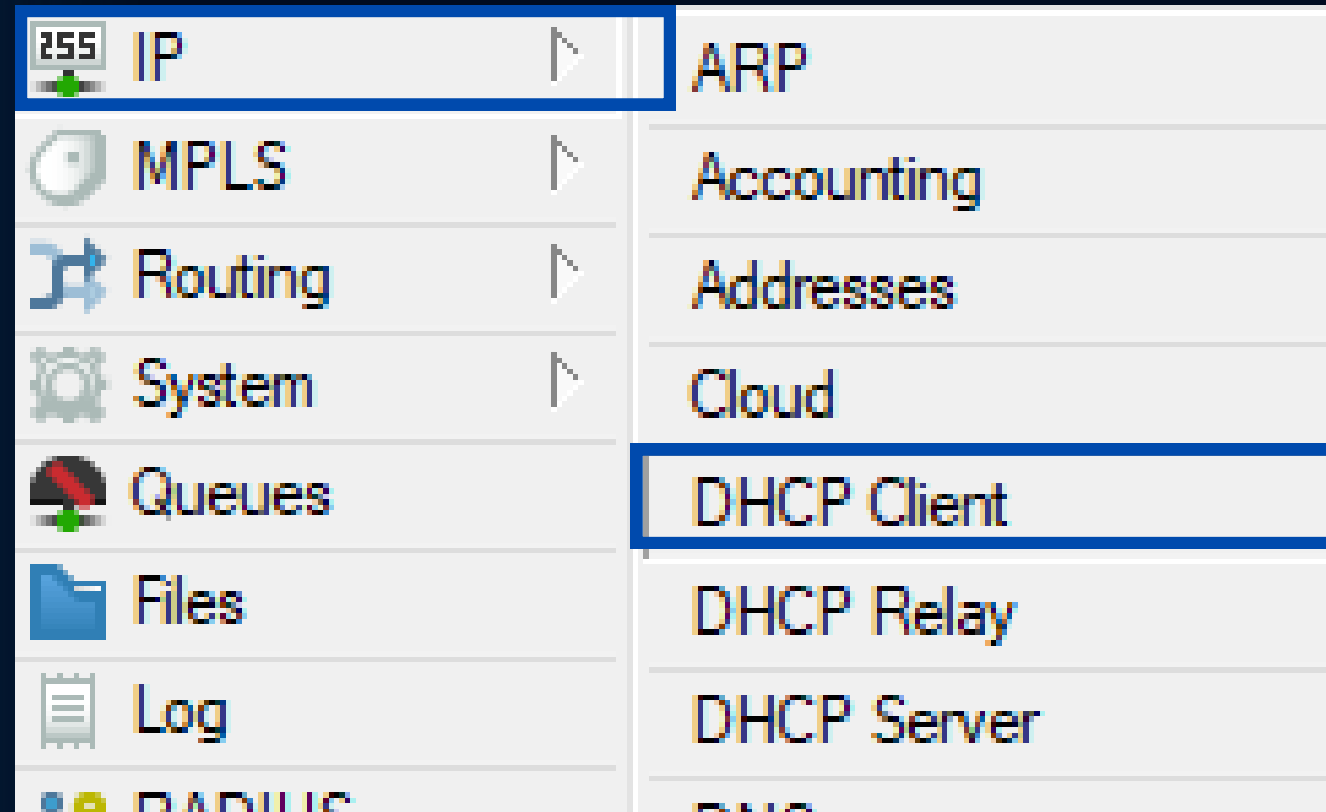
Kemudian klik "Apply"
lalu "OK"

BUKA NEW WINBOX KEMUDIAN LOGIN MENGGUNAKAN “USER”



HAPUS USER LIST “ADMIN”

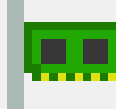
TAMBAHKAN DHCP CLIENT PADA ETHER 1 INTERNET



BUKA INTERFACES

 Quick Set

 CAPsMAN

 Interfaces

 Wireless

 Bridge

 PPP

 Switch

 Mesh

 IP



Wireless Tables

WiFi Interfaces W60G Station Nstreme Dual Access List Registration Connect List Security Profiles Channels Interworking Profiles

+ - ✓ ✗ [Icon] CAP WPS Client Setup Repeater Scanner Freq. Usage Alignment Wireless Sniffer Wireless Snooper Find

	Name	Type	Actual MTU	Tx	Rx	Tx Packet (p/s)	Rx Packet (p/s)	FP Tx
X	wlan1	Wireless (Atheros AR9...	1500	0 bps	0 bps	0	0	

KLIK WLAN1 DAN KLIK TANDA CENTANG
 DAN
 WLAN1 AKAN AKTIF

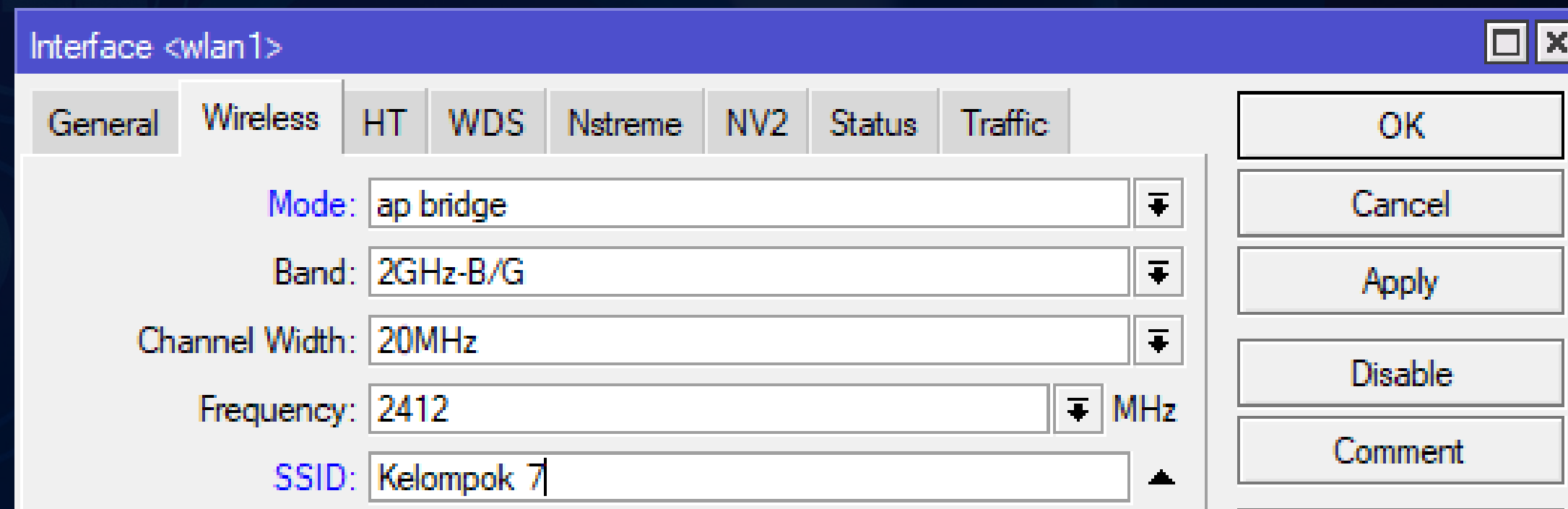
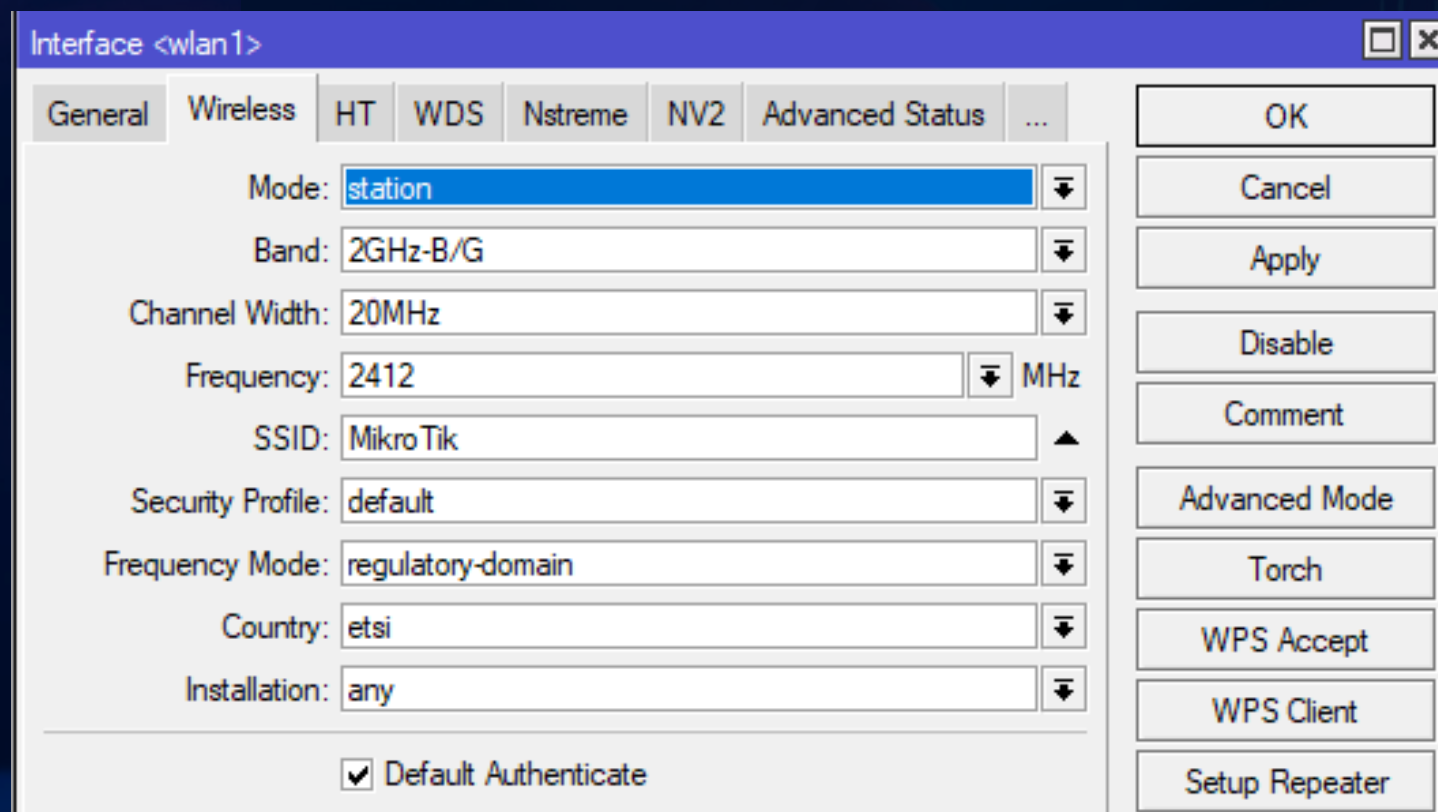
Wireless Tables

WiFi Interfaces W60G Station Nstreme Dual Access List Registration Connect List Security Profiles Channels Interworking Profiles

+ - [Checked] ✗ [Icon] CAP WPS Client Setup Repeater Scanner Freq. Usage Alignment Wireless Sniffer Wireless Snooper Find

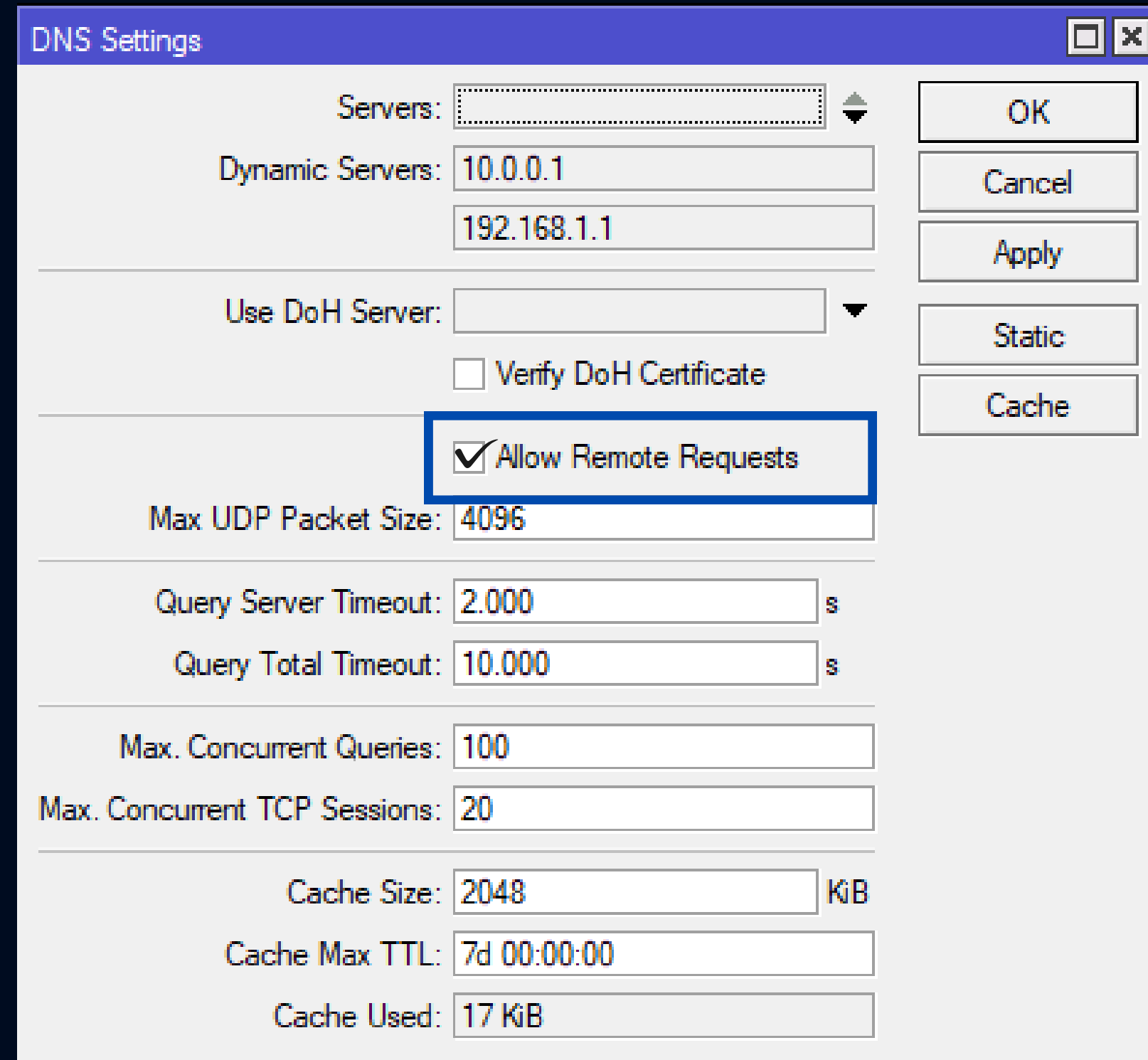
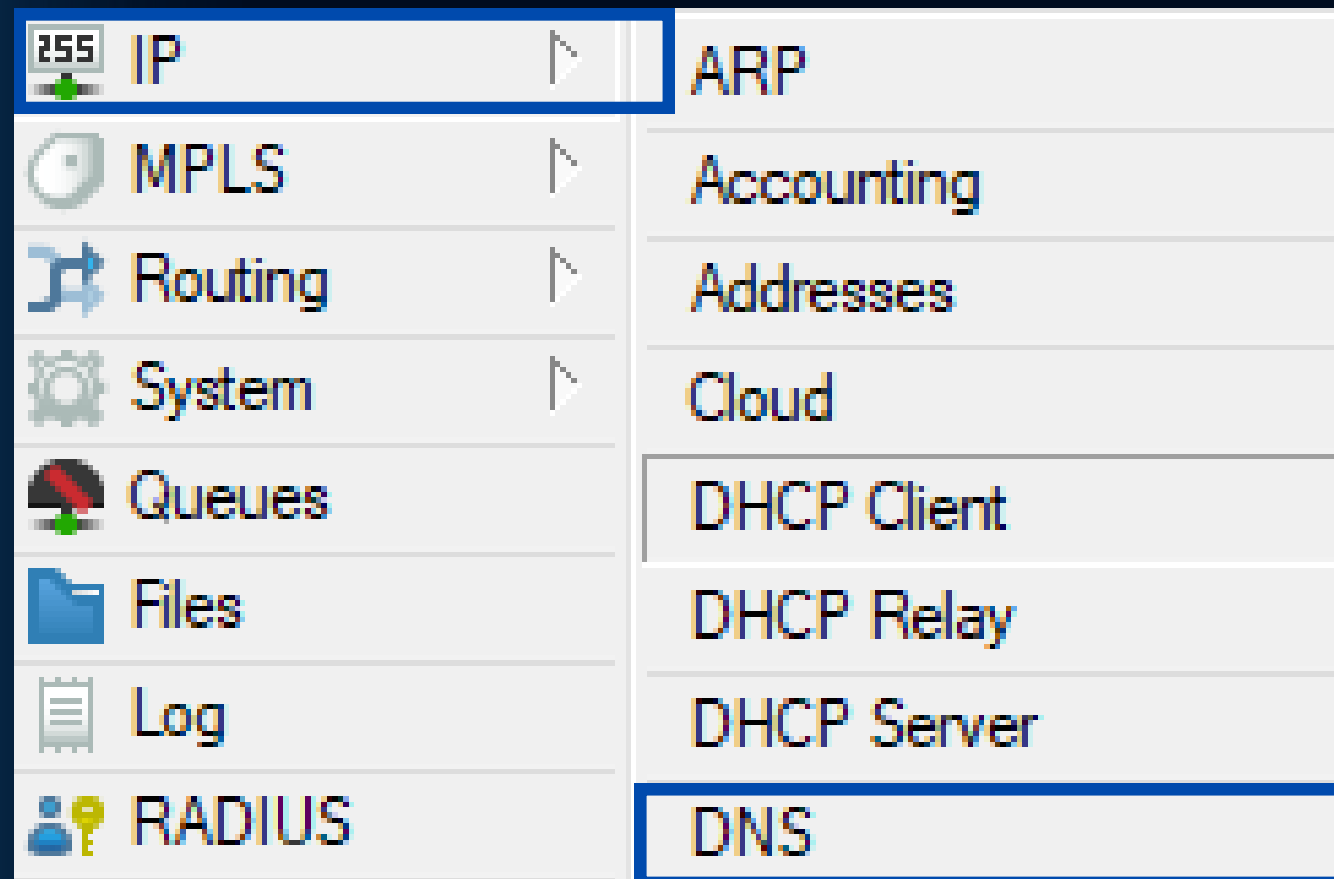
	Name	Type	Actual MTU	Tx	Rx	Tx Packet (p/s)	Rx Packet (p/s)	FP Tx
	wlan1	Wireless (Atheros AR9...	1500	0 bps	0 bps	0	0	

KLIK 2X WLAN2 KEMUDIAN GANTI MODE DAN SSID

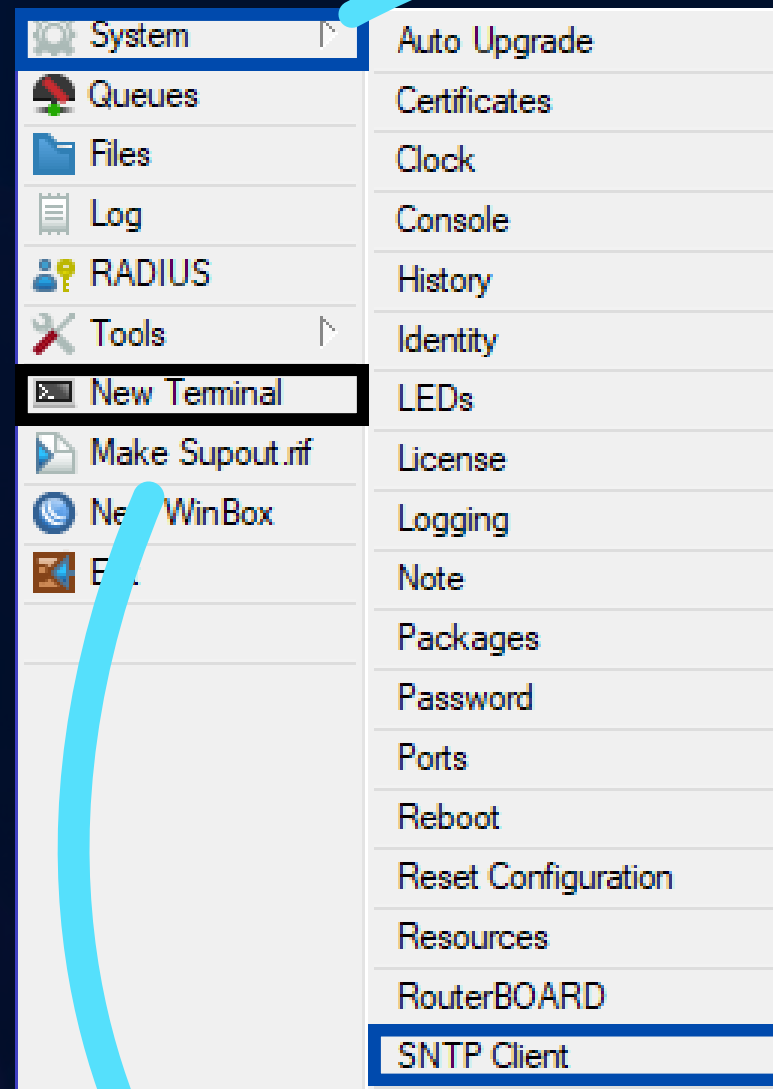


**MODE DI GANTI DENGAN AP BRIDGE
DAN SSID DIGANTI DENGAN NAMA SESUAI KELOMPOK**

MASUK KE IP LALU KE DNS CENTANG “ALLOW REMOTE REQUEST”



SETTING NTP SERVER



A screenshot of the SNTP Client configuration window. The 'Enabled' checkbox is checked. The 'Mode' is set to 'unicast'. The 'Primary NTP Server' is set to '202.4.186.250' and the 'Secondary NTP Server' is set to '116.12.47.30'. The 'Server DNS Names' and 'Dynamic Servers' fields are empty. The 'Poll Interval' is set to '32 s'. The 'Active Server' is set to '202.4.186.250'. The 'Last Update From' is set to '202.4.186.250'. The 'Last Update' is set to '00:00:02 ago'. The 'Last Adjustment' is set to '22 701 us'. The 'Last Bad Packet From', 'Last Bad Packet', and 'Last Bad Packet Reason' fields are empty. The 'OK', 'Cancel', and 'Apply' buttons are on the right.

```
[user@Kelompok 7] > ping 0.id.pool.ntp.org
SEQ HOST
0 202.4.186.250 Primary NTP Server
1 202.4.186.250
2 202.4.186.250
```

```
[user@Kelompok 7] > ping 1.id.pool.ntp.org
SEQ HOST
0 116.12.47.30 Secondary NTP Server
1 116.12.47.30
2 116.12.47.30
```


ATUR WAKTU SESUAI DAERAH MASING-MASING

Clock

Time **Manual Time Zone**

Time: 12:56:48

Date: Feb/28/2024

☒ Time Zone Autodetect

Time Zone Name: Asia/Makassar

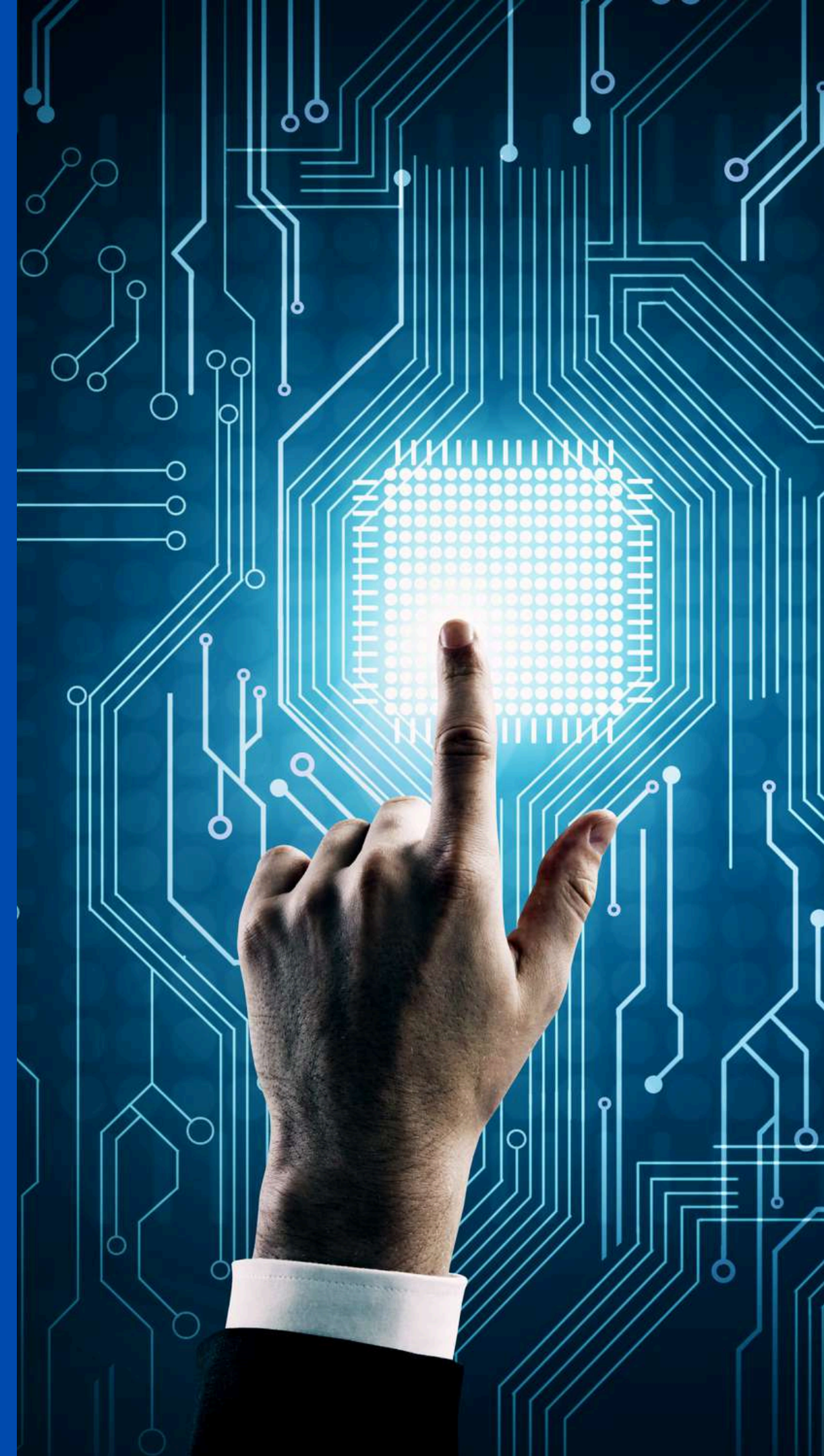
GMT Offset: +08:00

☐ DST Active

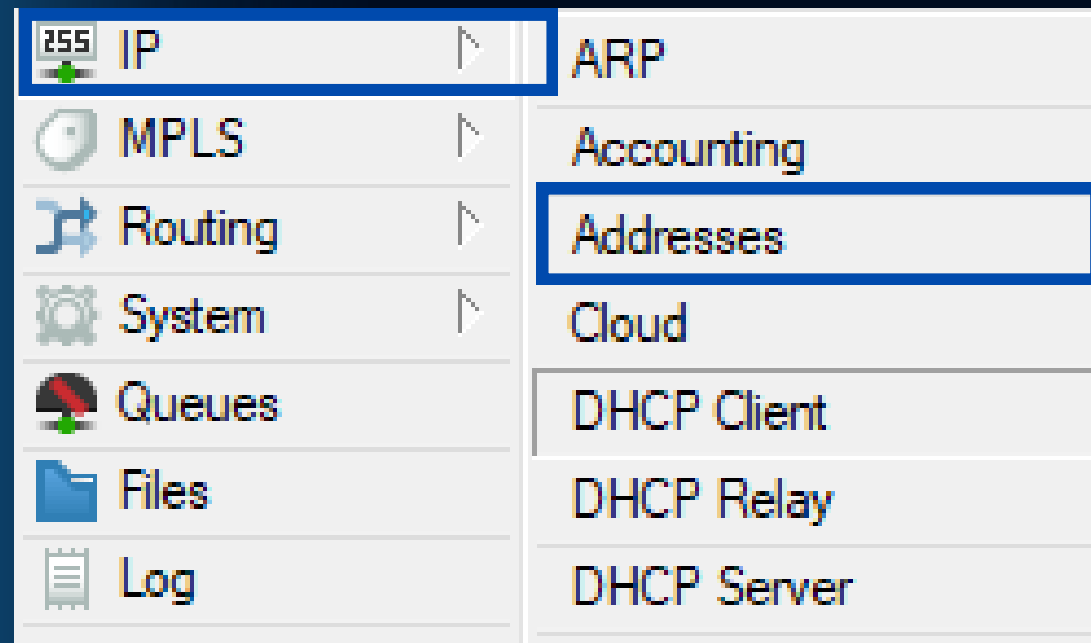
OK

Cancel

Apply



BERIKAN IP PADA INTERFACE ETHER 2 DAN WLAN1.



Address: 192.168.70.1/25

Network: 192.168.70.0

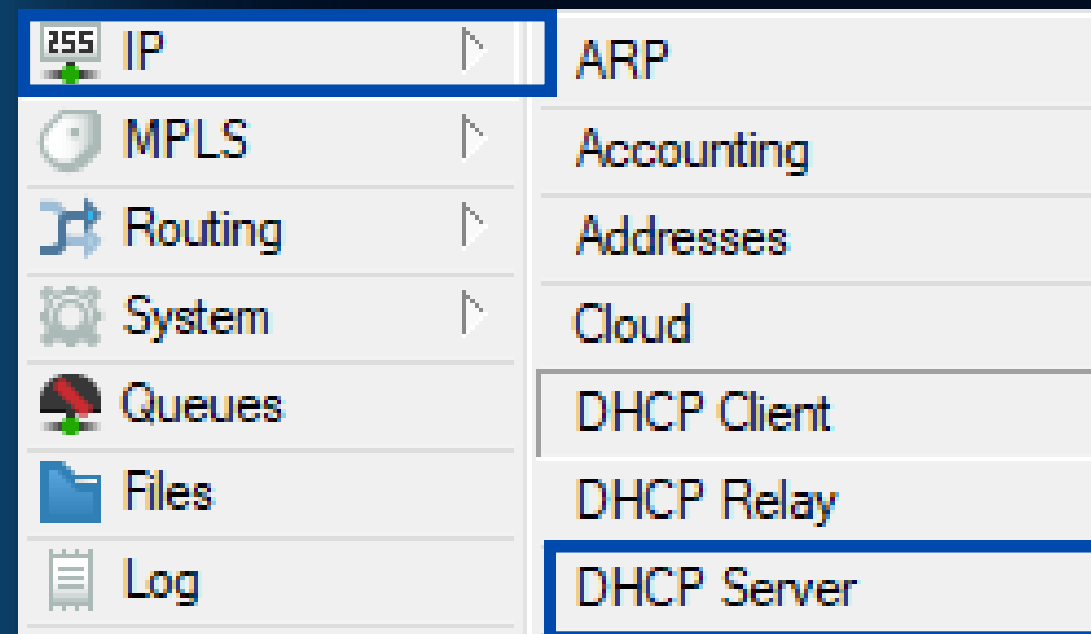
Interface: ether2-LOKAL

Address: 192.168.80.1/24

Network: 192.168.80.0

Interface: wlan1-WIRELESS

MASUK PADA DHCP SERVER KEMUDIAN KLIK “DHCP SETUP”



DHCP Server

DHCP Networks Leases Options Option Sets Vendor Classes Alerts

+ - ✓ ✕ Filter DHCP Config DHCP Setup

Name	Interface	Relay	Lease Time	Address Pool
------	-----------	-------	------------	--------------

DHCP SETUP

DHCP Setup

Select interface to run DHCP server on

DHCP Server Interface: ether2-LOKAL

Back Next Cancel

DHCP Setup

Select pool of ip addresses given out by DHCP server

Addresses to Give Out: 68.70.2-192.168.70.100

Back Next Cancel

DHCP Setup

Select interface to run DHCP server on

DHCP Server Interface: wlan1-WIRELESS

Back Next Cancel

DHCP Setup

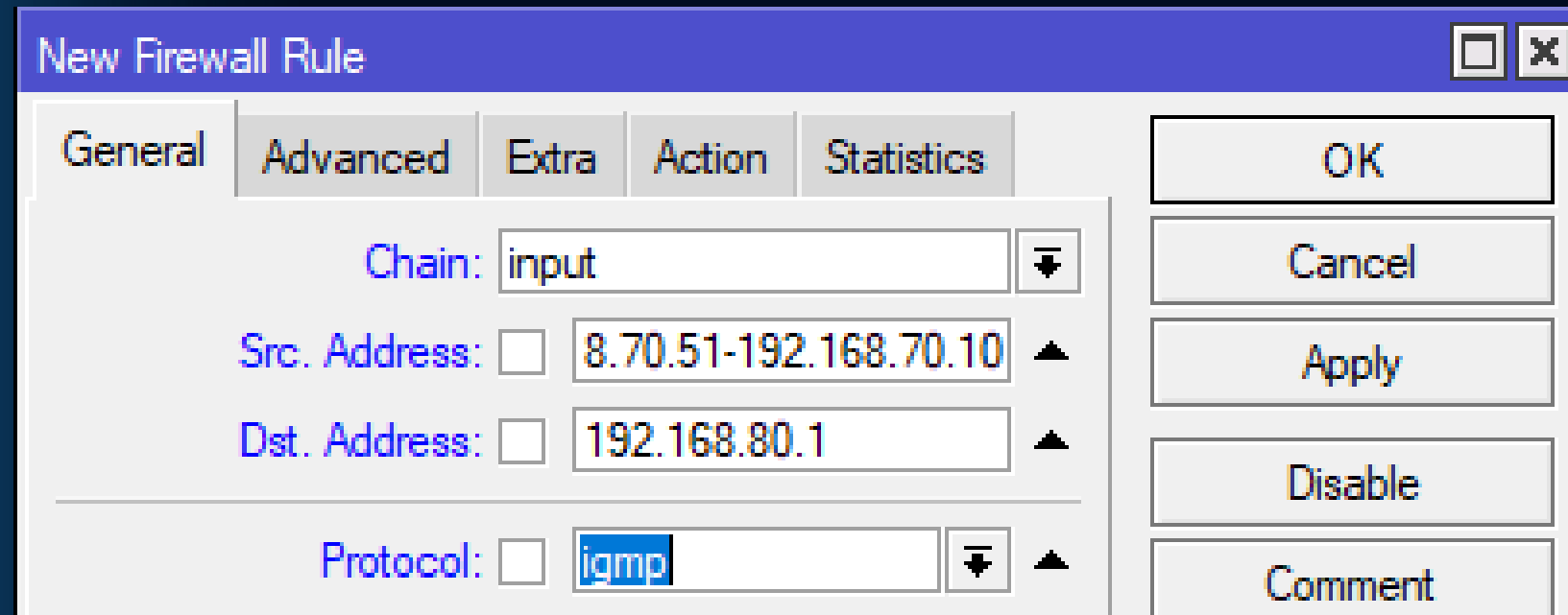
Select pool of ip addresses given out by DHCP server

Addresses to Give Out: 68.80.2-192.168.80.100

Back Next Cancel

**DHCP POOL
SEBANYAK 99
CLIENT**

**Buat firewall agar IP 192.168.70.51-192.168.70.100
tidak dapat ping ke client wireless**



Keterangan:

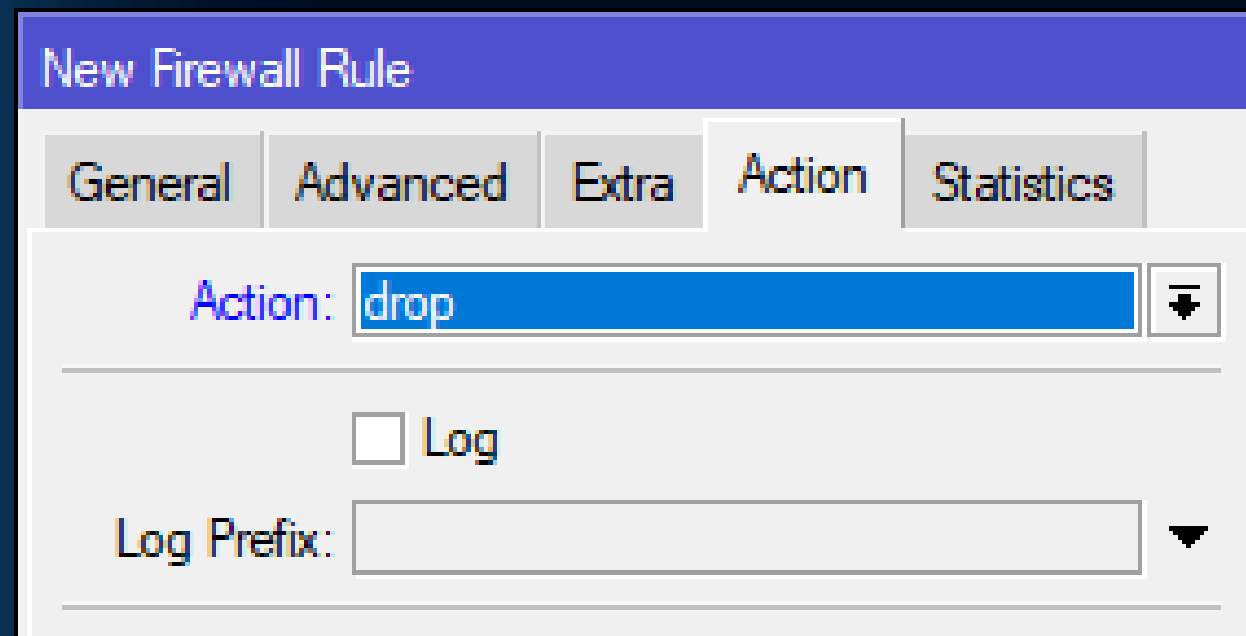
Chain input

Src Address 192.168.70.51-192.168.70.100

Dst Address IP Client Wireless 192.168.80.1

Protocol icmp

CLIENT WIRELESS : 192.168.80.1



Action drop

```
C:\WINDOWS\system32\cmd. X + v

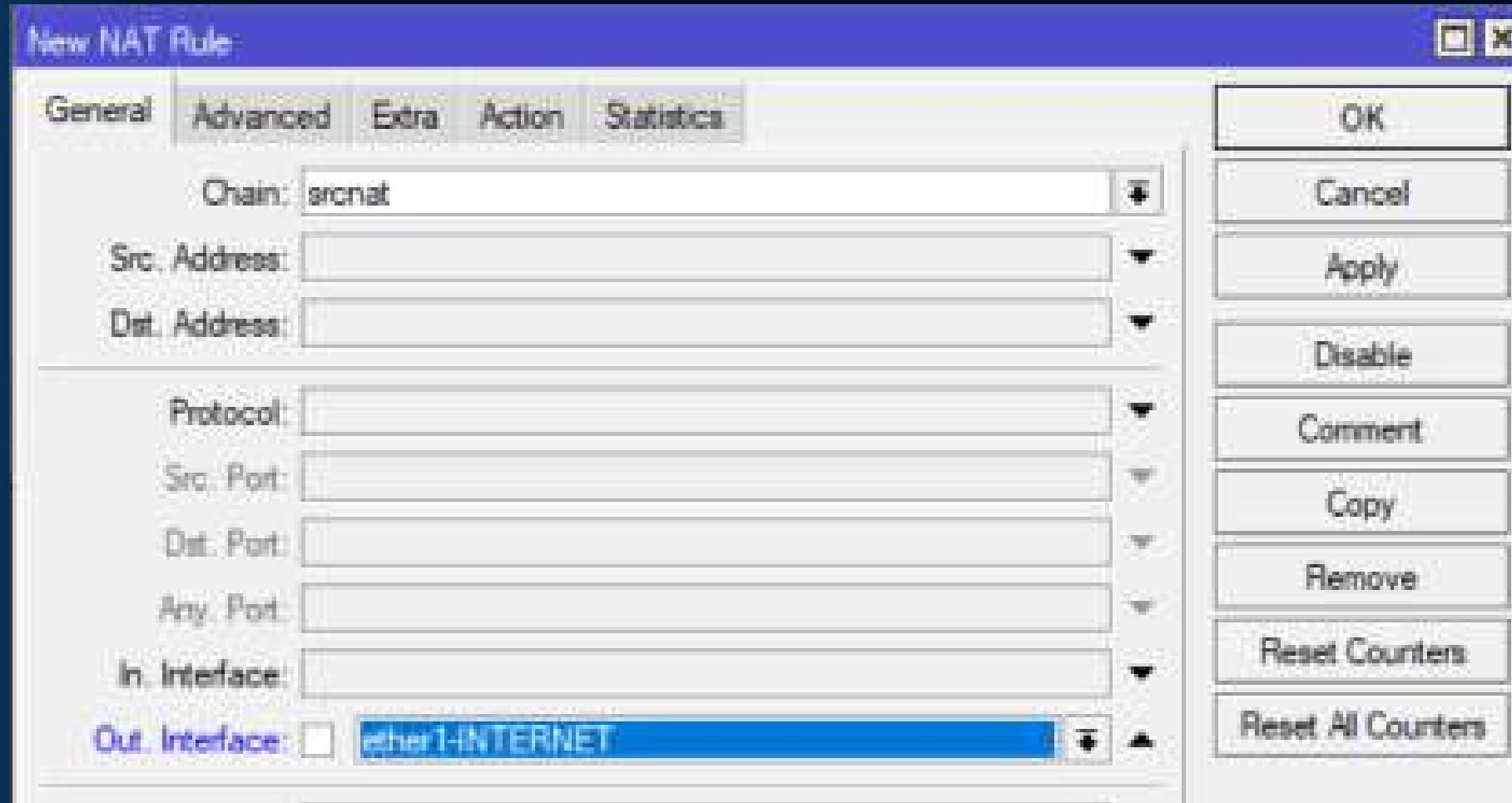
Microsoft Windows [Version 10.0.22621.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>ping 192.168.80.1

Pinging 192.168.80.1 with 32 bytes of data:
Request timed out.
```

**Ping IP client wireless jika tidak reply
berarti firewall berhasil**

Pergi ke Firewall Nat Rule lalu pada Out.Interface pilih ether1-INTERNET.

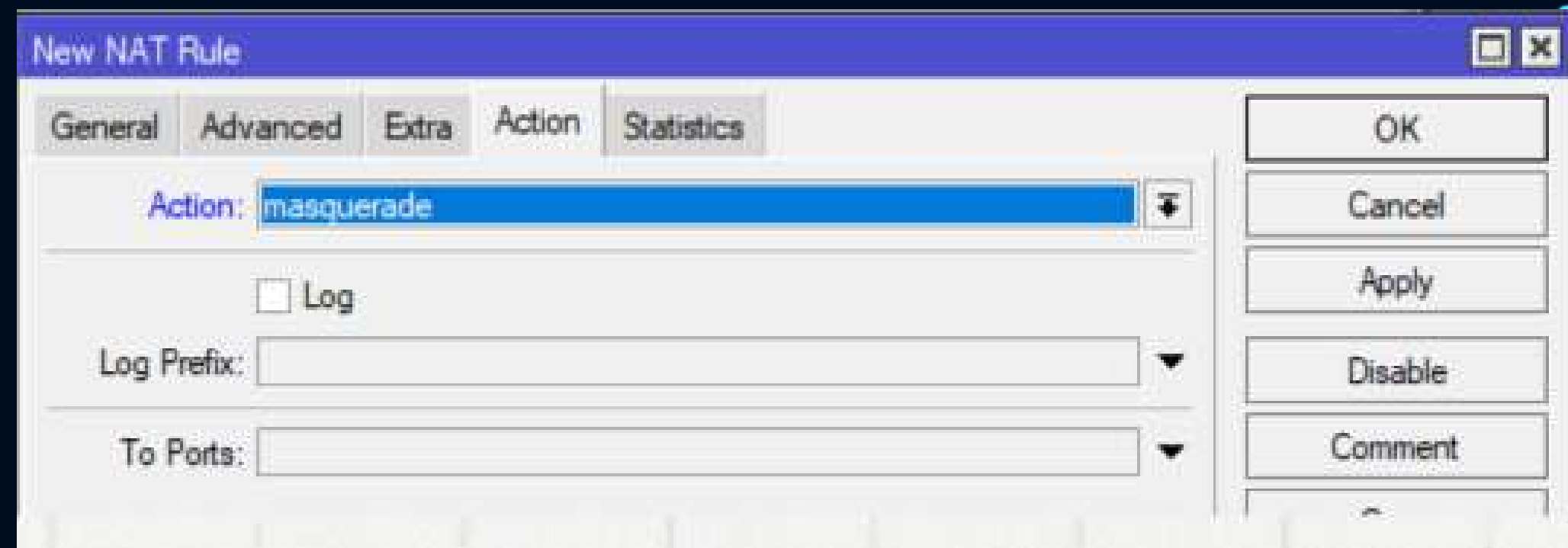


The screenshot shows the 'New NAT Rule' dialog box with the 'General' tab selected. The 'Chain' is set to 'srcnat'. The 'Out. Interface' is set to 'ether1-INTERNET'. The 'Action' tab is also visible, showing the 'masquerade' action selected.

Field	Value
Chain	srcnat
Src. Address	
Dst. Address	
Protocol	
Src. Port	
Dst. Port	
Any. Port	
In. Interface	
Out. Interface	ether1-INTERNET

Buttons: OK, Cancel, Apply, Disable, Comment, Copy, Remove, Reset Counters, Reset All Counters.

Kemudian klik “masquerade”

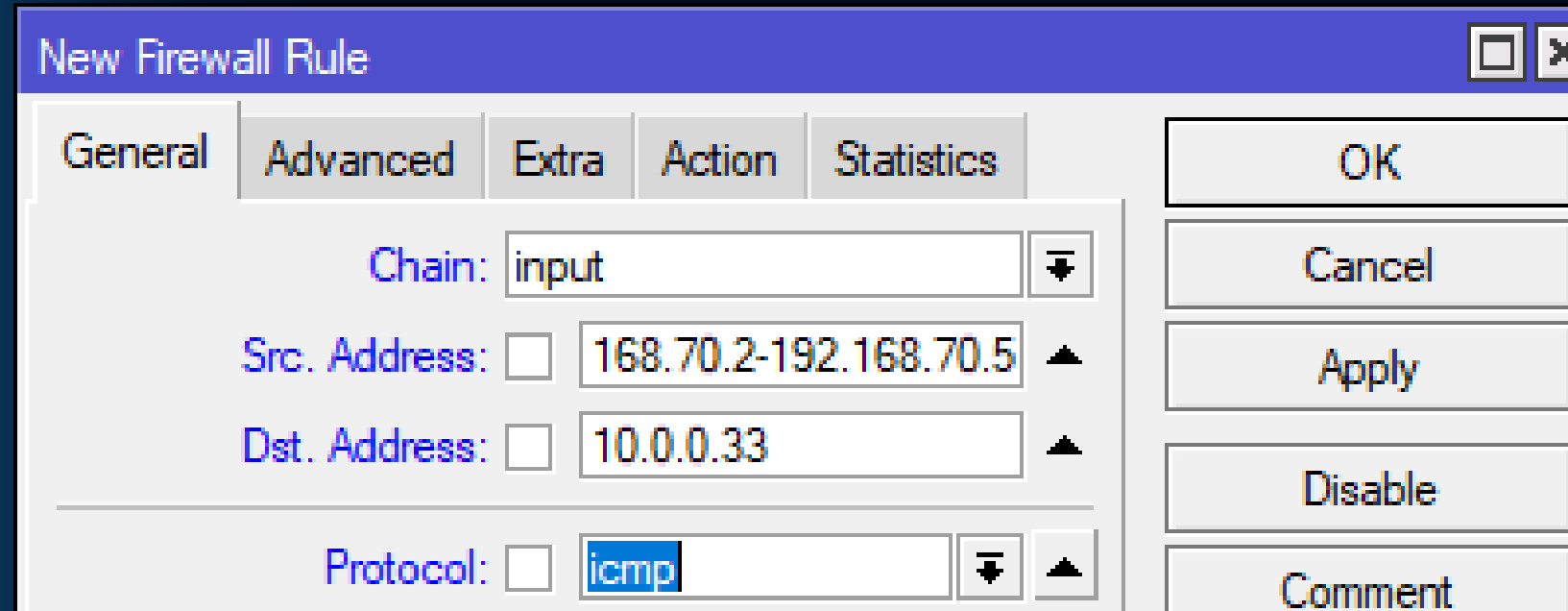


The screenshot shows the 'New NAT Rule' dialog box with the 'Action' tab selected. The 'Action' is set to 'masquerade'. The 'Log' checkbox is unchecked. The 'Log Prefix' and 'To Ports' fields are empty.

Field	Value
Action	masquerade
Log	☐
Log Prefix	
To Ports	

Buttons: OK, Cancel, Apply, Disable, Comment, Copy, Remove, Reset Counters, Reset All Counters.

Buat firewall agar IP 192.168.70.2-192.168.70.50 tidak dapat ping ke router



New Firewall Rule

General Advanced Extra Action Statistics

Chain: input

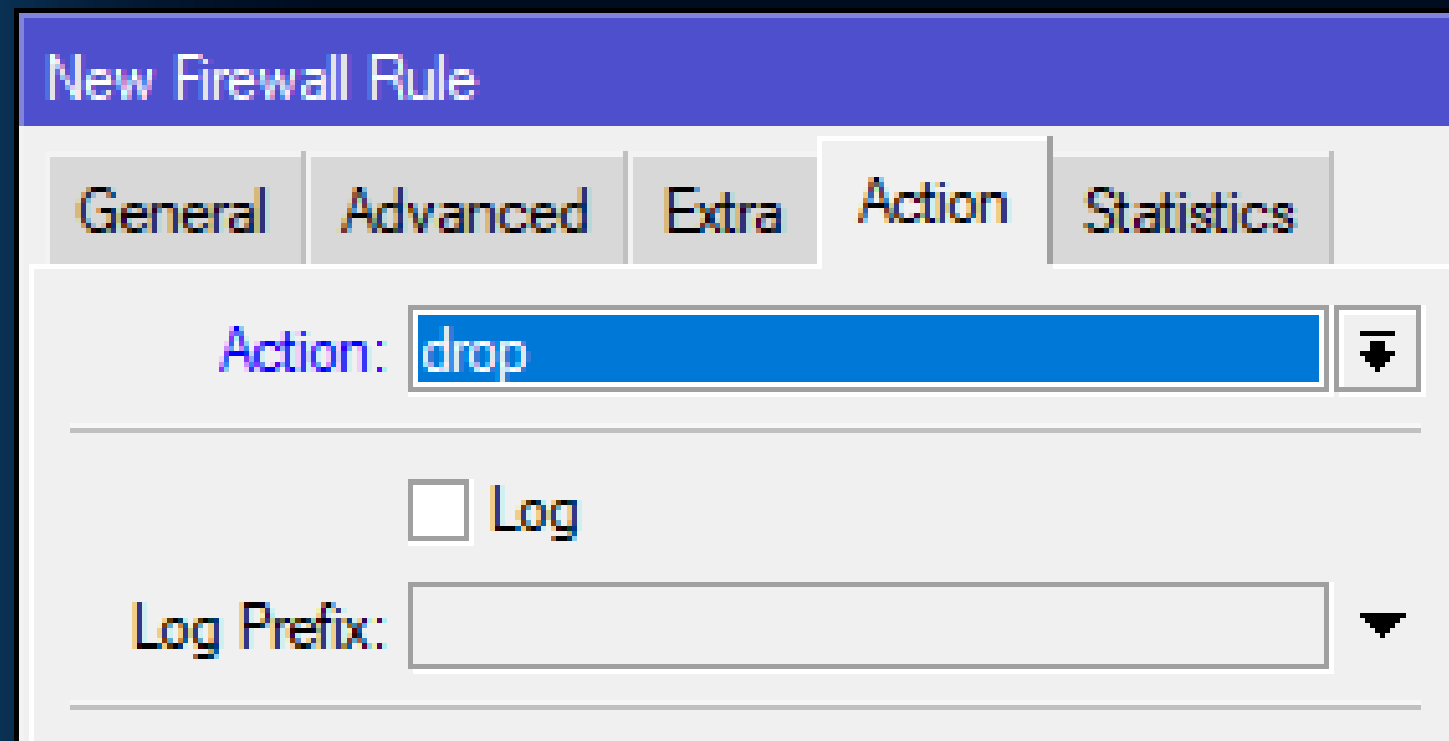
Src. Address: ☐ 168.70.2-192.168.70.5

Dst. Address: ☐ 10.0.0.33

Protocol: ☐ icmp

OK Cancel Apply Disable Comment

Keterangan:
Chain input
Src Address 192.168.70.2-192.168.70.50
Dst Address router 10.0.0.33
Protocol icmp



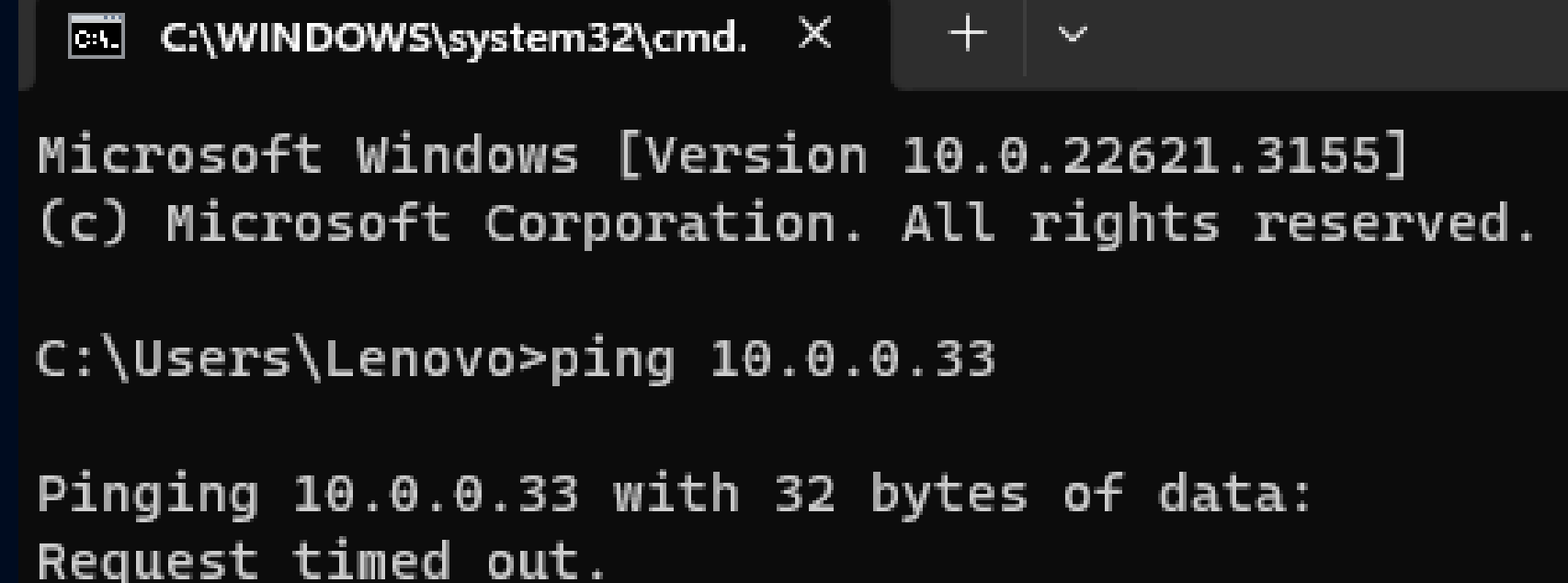
New Firewall Rule

General Advanced Extra Action Statistics

Action: drop

☒ Log

Log Prefix:



```
C:\WINDOWS\system32\cmd. X + v

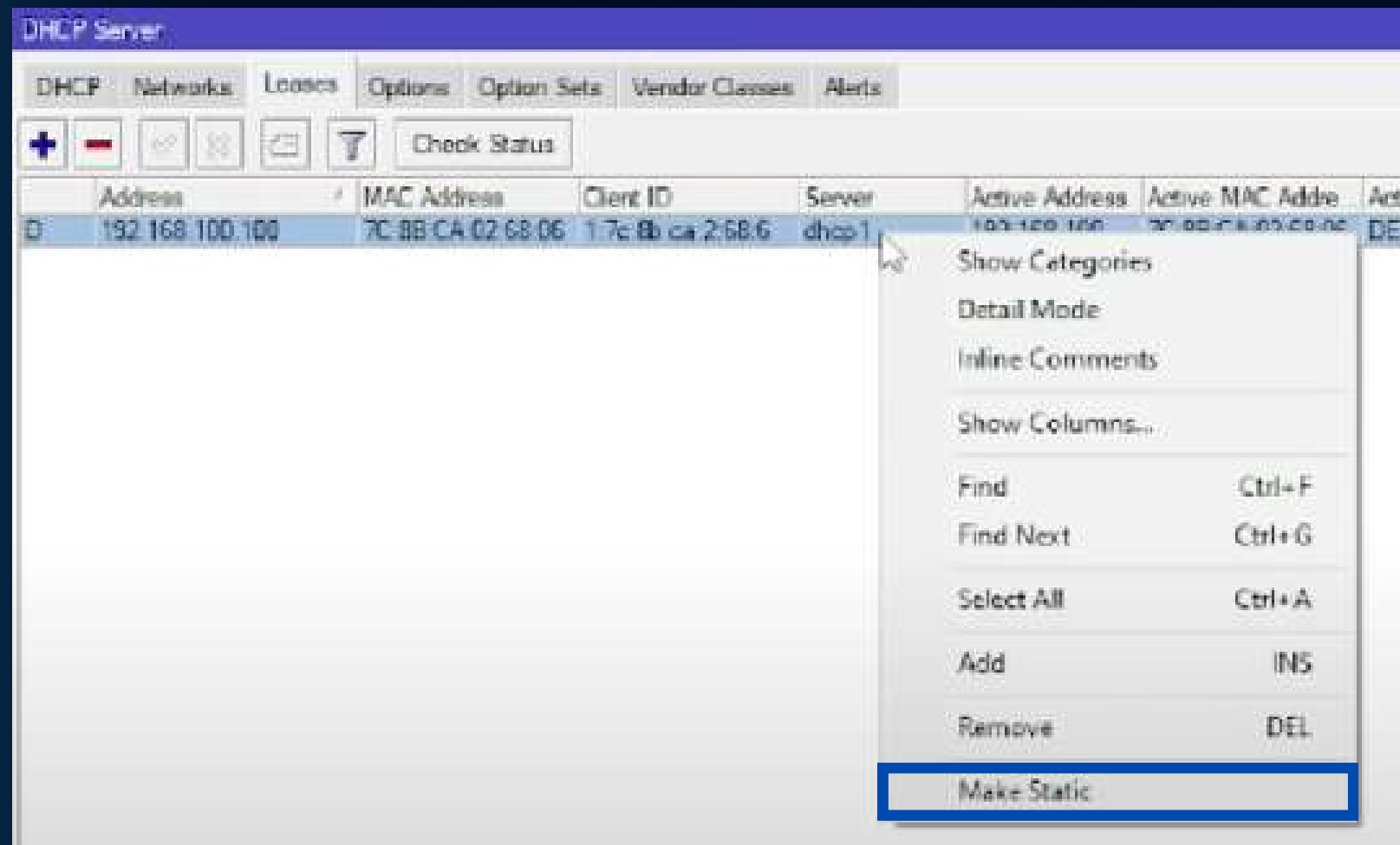
Microsoft Windows [Version 10.0.22621.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Lenovo>ping 10.0.0.33

Pinging 10.0.0.33 with 32 bytes of data:
Request timed out.
```

Ping IP router jika tidak reply berarti firewall berhasil

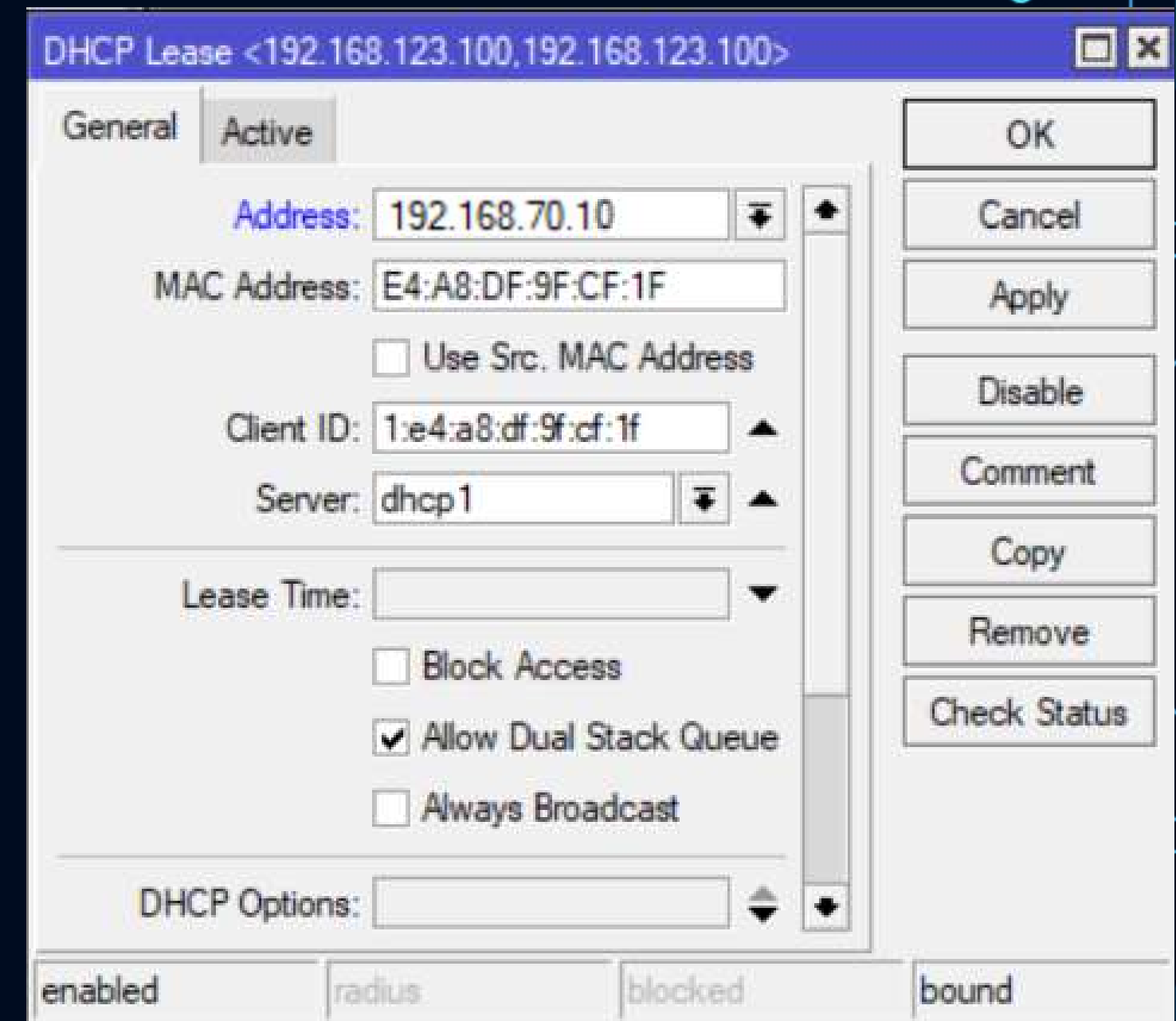
Buat IP DHCP Server menjadi Static



```
C:\Users\Lenovo>ping 10.0.0.33
```

```
Pinging 10.0.0.33 with 32 bytes of data:  
Request timed out.
```

Ubah IP 192.168.70.100 menjadi 192.168.70.10



Tes ping IP Router, jika tidak reply berarti sudah berhasil

Buat rule agar setiap akses ke router tercatat di logging dan tersimpan di disk.

Log				
Freeze				
#	Time	Buffer	Topics	Me
28	Feb/28/2024 12:49:44	memory	system, info, account	use
29	Feb/28/2024 12:50:06	memory	system, info	use
30	Feb/28/2024 12:51:34	memory	system, info	dns
31	Feb/28/2024 12:53:41	memory	system, info, account	use
32	Feb/28/2024 12:55:50	memory	system, info	SN
33	Feb/28/2024 12:55:57	memory	dhcp, info	dhc

Log Rule <critical>

Topics: ☐ critical

Prefix:

Action: disk

OK

Cancel

Apply

Logging				
RulesActions Find				
	Topics	Prefix	Action	
*	critical		echo	
*	error		memory	
*	info		memory	
*	waming		memory	

Logging				
RulesActions Find				
	Topics	Prefix	Action	
*	critical		disk	
*	error		disk	
*	info		disk	
*	waming		disk	

Pergi ke firewall – filter rules – chain input lalu general – action – log – log Prefix : Tersimpan di disk

New Firewall Rule

General Advanced Extra Action Statistics

Chain:

Src. Address:

Dest. Address:

OK

Cancel

Apply

Disable

New Firewall Rule

General Advanced Extra Action Statistics

Action:

☐ Log

Log Prefix:

Log

#	Time	Buffer	Topics	Message
975	Feb/28/2024 13:21:50	disk	firewall, info	Tersimpan
976	Feb/28/2024 13:21:50	disk	firewall, info	Tersimpan
977	Feb/28/2024 13:21:50	disk	firewall, info	Tersimpan
978	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
979	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
980	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
981	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
982	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
983	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
984	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
985	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
986	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
987	Feb/28/2024 13:21:51	disk	firewall, info	Tersimpan
988	Feb/28/2024 13:21:52	disk	firewall, info	Tersimpan
989	Feb/28/2024 13:21:52	disk	firewall, info	Tersimpan
990	Feb/28/2024 13:21:52	disk	firewall, info	Tersimpan

buffer berhasil diubah ke “disk”

Ganti SSID pada jaringan wireless

Wireless Tables

WiFi InterfacesW60G StationNstreme DualAccess ListRegistrationConnect ListSecurity ProfilesChannelsInterworking Profiles

+CAPWPS ClientSetup RepeaterScannerFreq. UsageAlignmentWireless SnifferWireless SnooperFind

	Name	Type	Actual MTU	Tx	Rx	Tx Packet (p/s)	Rx Packet (p/s)	FP Tx
R	wlan1	Wireless (Atheros AR9...	1500	5.3 kbps	3.4 kbps	11	4	

Interface <wlan1-WIRELESS>

GeneralWirelessHTHT MCSWDSNstremeNV2Status...

Mode:ap bridge

Band:2GHz-B/G/N

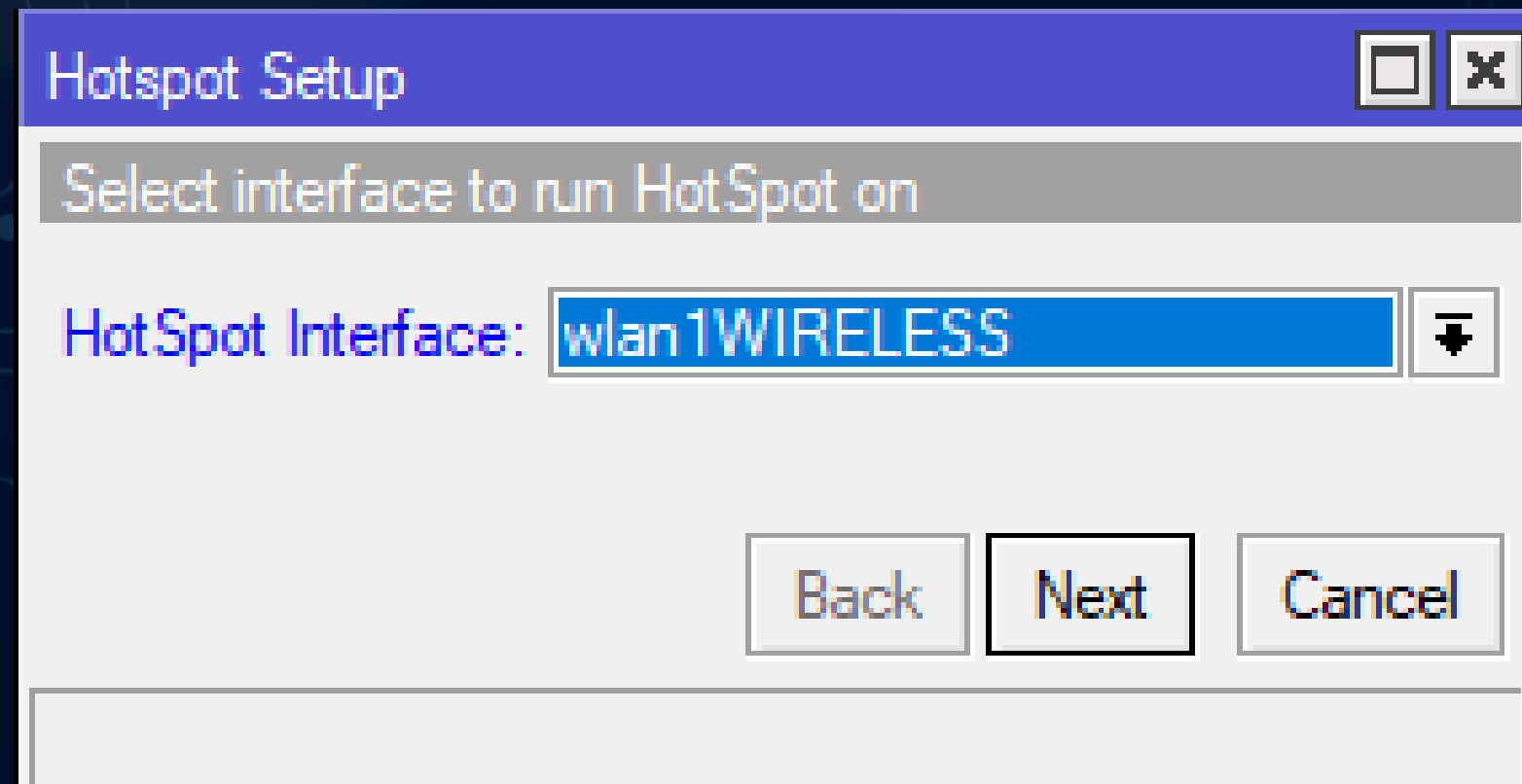
Channel Width:20MHz

Frequency:2412 MHz

SSID:Kelompok 7

Keterangan:
Mode : ap bridge
Band : 2GHz-B/G/N
SSID : Kelompok 7

Hotspot Setup

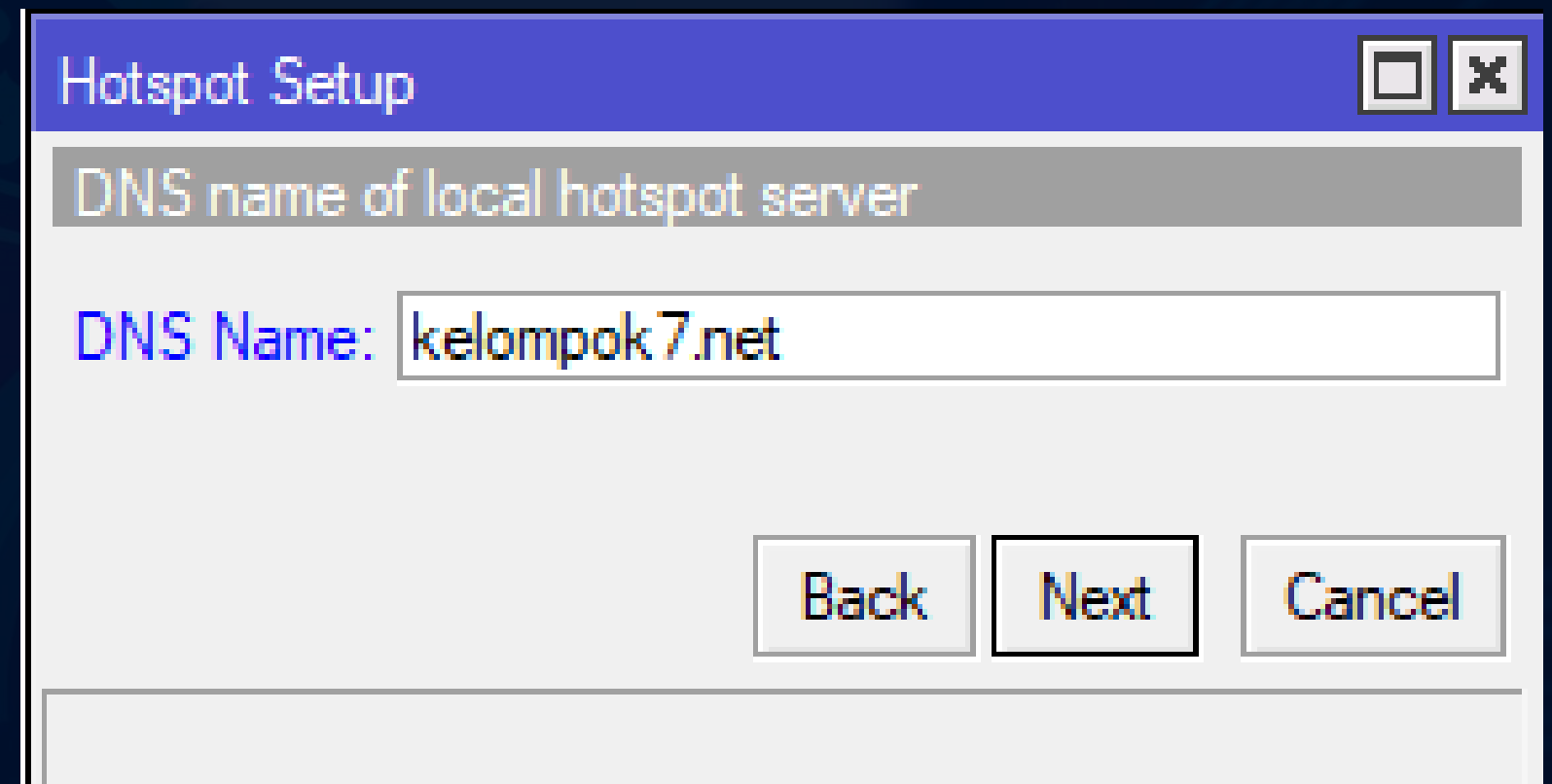


Hotspot Setup

Select interface to run HotSpot on

HotSpot Interface: wlan1WIRELESS

Back Next Cancel

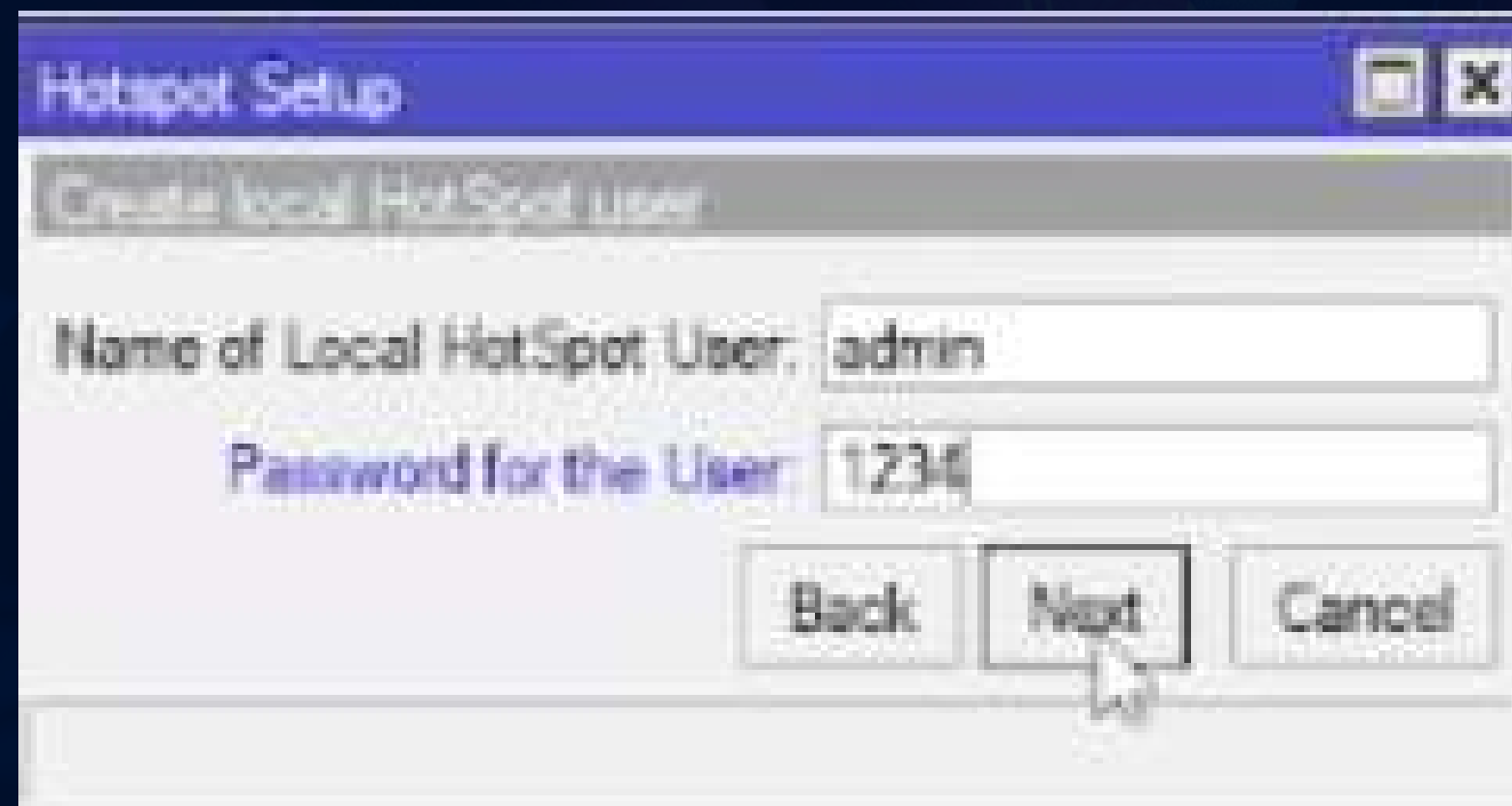


Hotspot Setup

DNS name of local hotspot server

DNS Name: kelompok7.net

Back Next Cancel



Hotspot Setup

Create local HotSpot user

Name of Local HotSpot User: admin

Password for the User: 1234

Back Next Cancel

Keterangan:
Hotspot Interface : wlan1-WIRELESS
DNS Name : kelompok7.net
User : admin
Password : 1234

Hotspot Setup

Hotspot

Servers Server Profiles Users User Profiles Active Hosts IP Bindings Services

+ - Filter

Name	DNS Name	HTML Directory	Rate Limit (rx/tx)
default		hotspot	
hsprof1	kelompok7.net	hotspot	

Hotspot Server Profile <hsprof1>

General Login RADIUS

Login By: ☐ MAC ☒ Cookie

☒ HTTP CHAP ☐ HTTPS

☐ HTTP PAP ☐ Trial

☐ MAC Cookie

Hotspot Server Profile <hsprof1>

General Login RADIUS

☒ Use RADIUS

Default Domain:

Location ID:

Location Name:

MAC Format: XX:XX:XX:XX:XX:XX

☒ Accounting

Interim Update:

NAS Port Type: 19 (wireless-802.11)

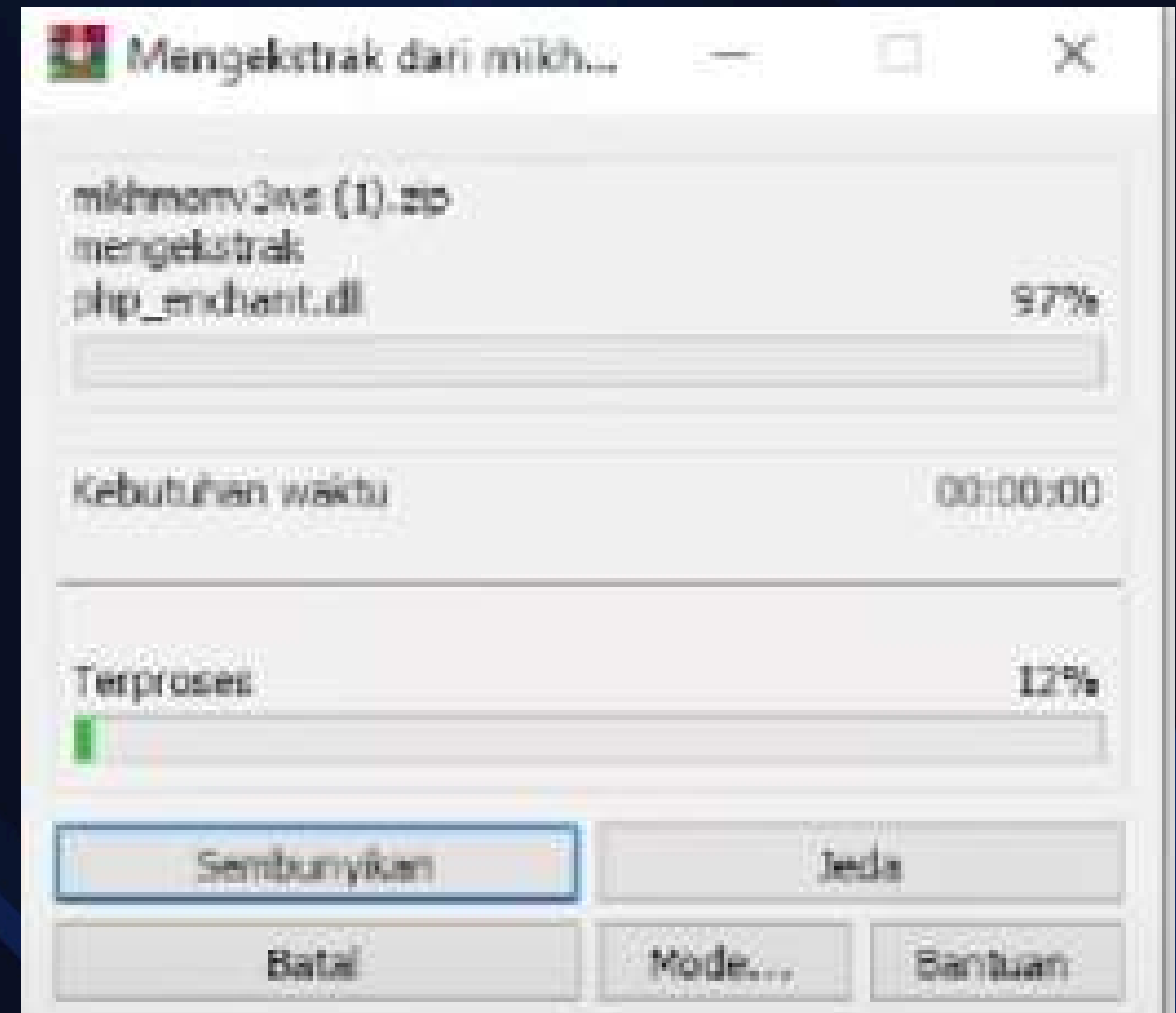
OK Cancel Apply Copy Remove

Keterangan:
Klik 2 kali pada kelompok7.net
Checklist Use Radius
Unchecklist Cookie

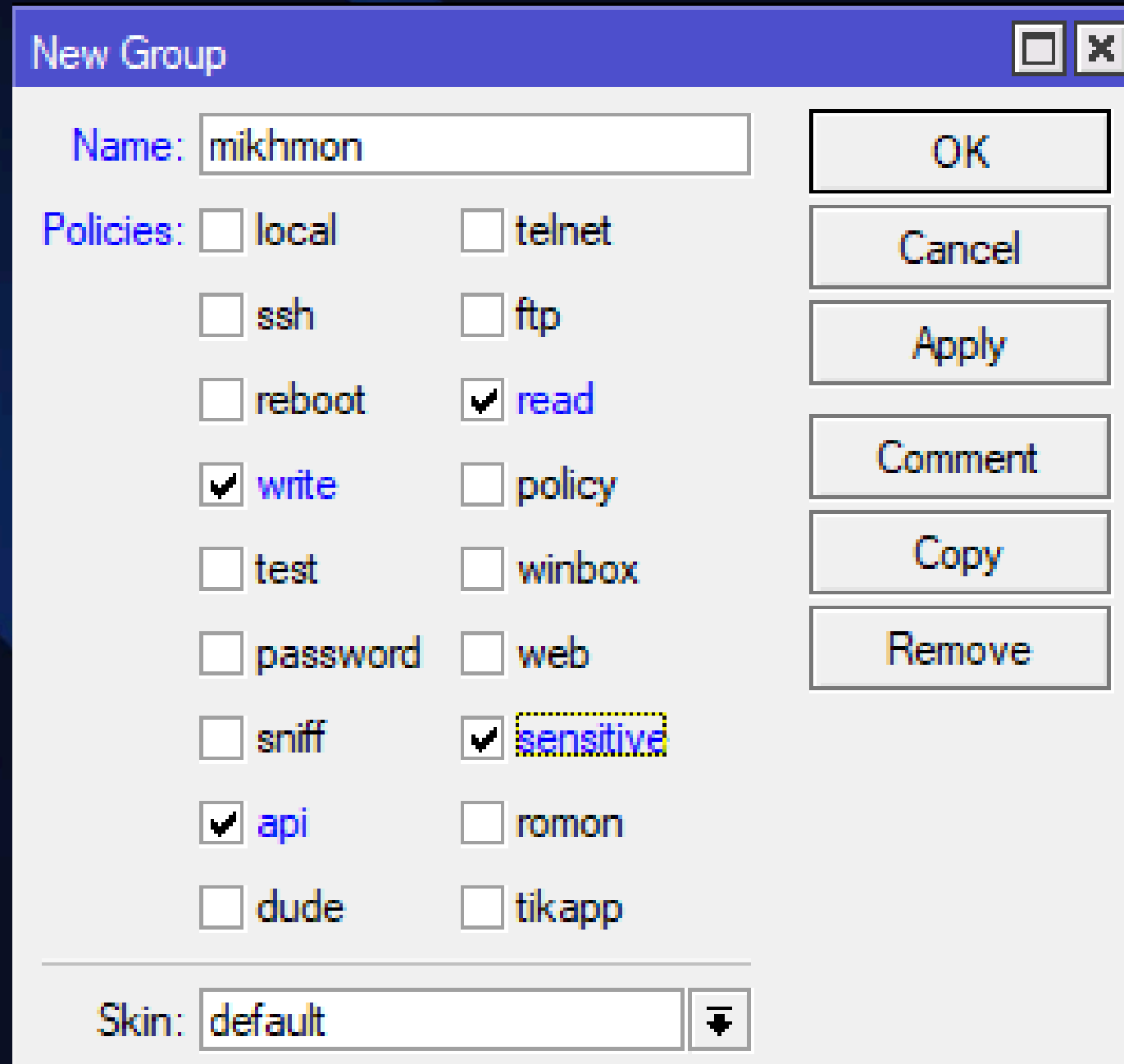
Download Mikhmon, kemudian extrack file



link download
<https://laksa19.github.io>



Pergi ke User list kemudian buat new group



New Group

Name: mikhmon

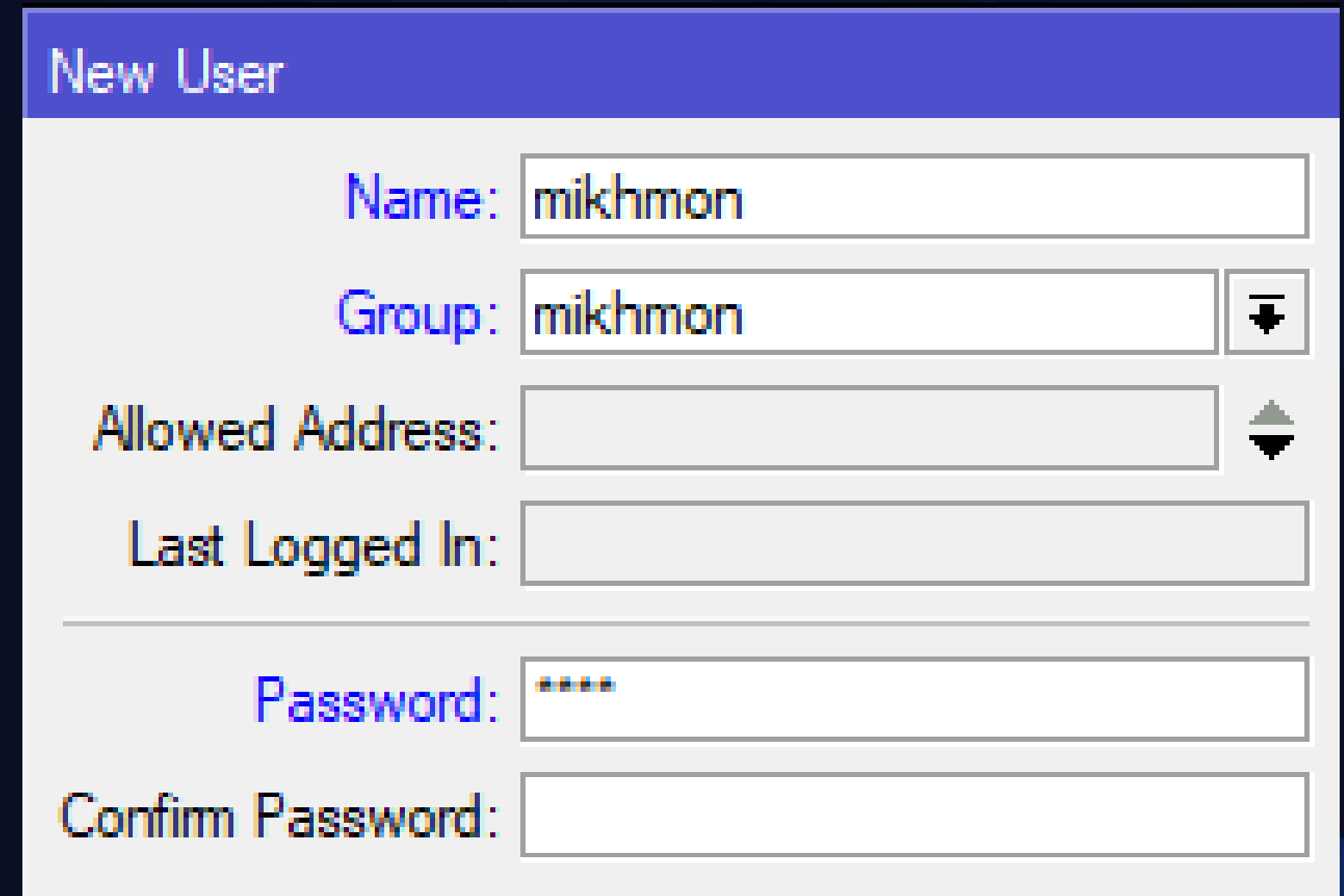
Policies:

☐ local	☐ telnet
☐ ssh	☐ ftp
☐ reboot	☒ read
☒ write	☐ policy
☐ test	☐ winbox
☐ password	☐ web
☐ sniff	☒ sensitive
☒ api	☐ romon
☐ dude	☐ tikapp

Skin: default

OK
Cancel
Apply
Comment
Copy
Remove

Name : mikhmon
Checklist : write, api, read
dan sensitive



New User

Name: mikhmon

Group: mikhmon

Allowed Address:

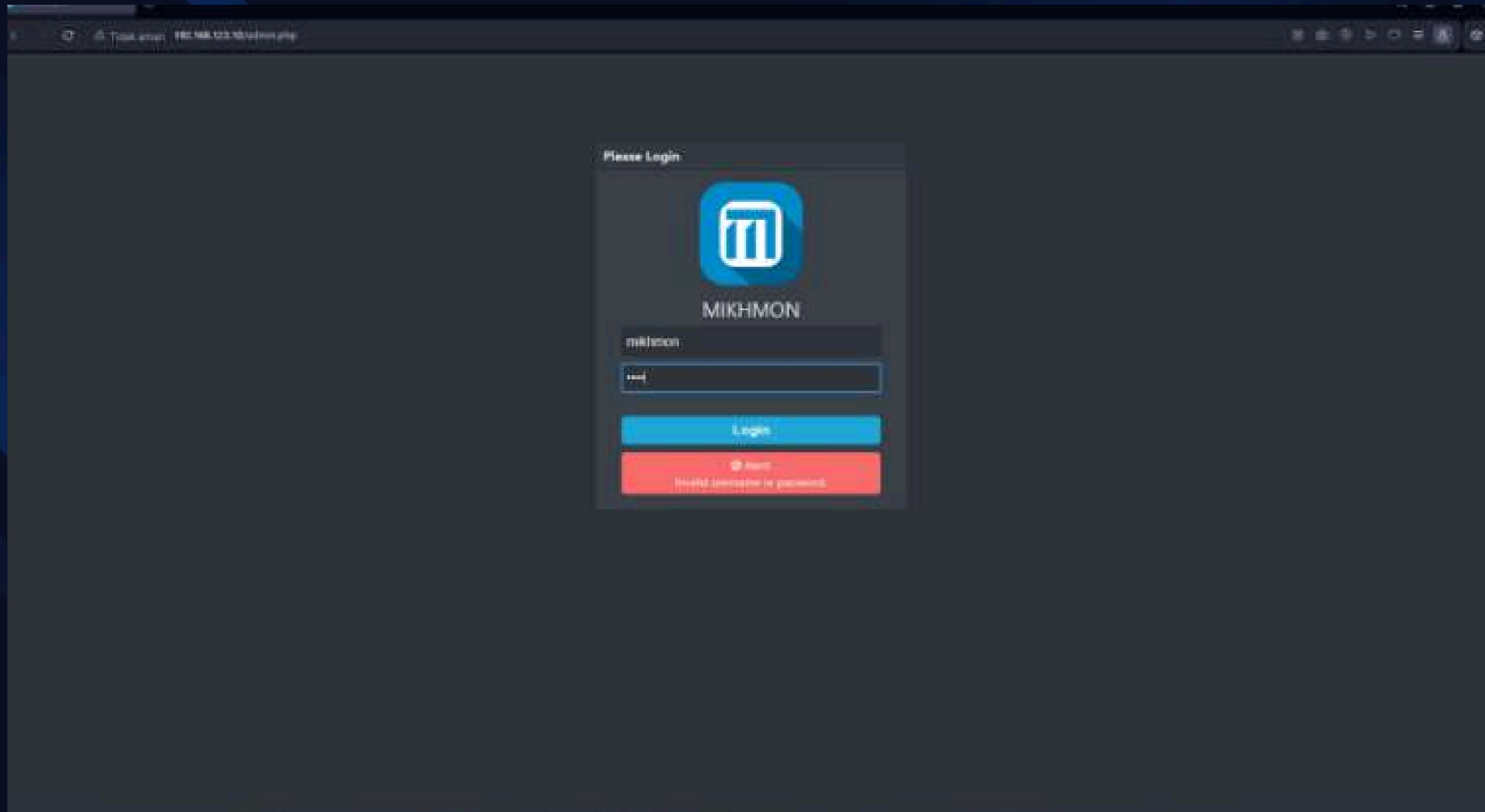
Last Logged In:

Password:

Confirm Password:

Name : mikhmon
Group : mikhmon
Password : 1234

Login ke mikhmon menggunakan user dan password yang telah disetting



**Setting seperti gambar di bawah.
Klik save, ping lalu connect.**

Ping Test [192.168.80.1:8728]

Host : 192.168.80.1 Port : 8728

Ping OK

 Close

**Setting seperti gambar di bawah.
Klik save lalu close.**

+ Add User Profile

✕ Close

💾 Save

Name	20-ORANG
Address Pool	none
Shared Users	1
Rate limit [up/down]	1M/1M
Expired Mode	Remove & Record
Validity	1d
Price Rp	5000
Selling Price Rp	5000
Lock User	Disable
Parent Queue	none

**Setting seperti gambar di bawah.
Klik generate lalu print.**

 **Generate User**

 Close

 User List

 Generate

 Print

 QR

 Small

Qty

20

Server

all

User Mode

Username & Password

Name Length

4

Prefix

Character

Random 5AB2C34D

Profile

20-ORANG

Time Limit

1d

Data Limit

MB

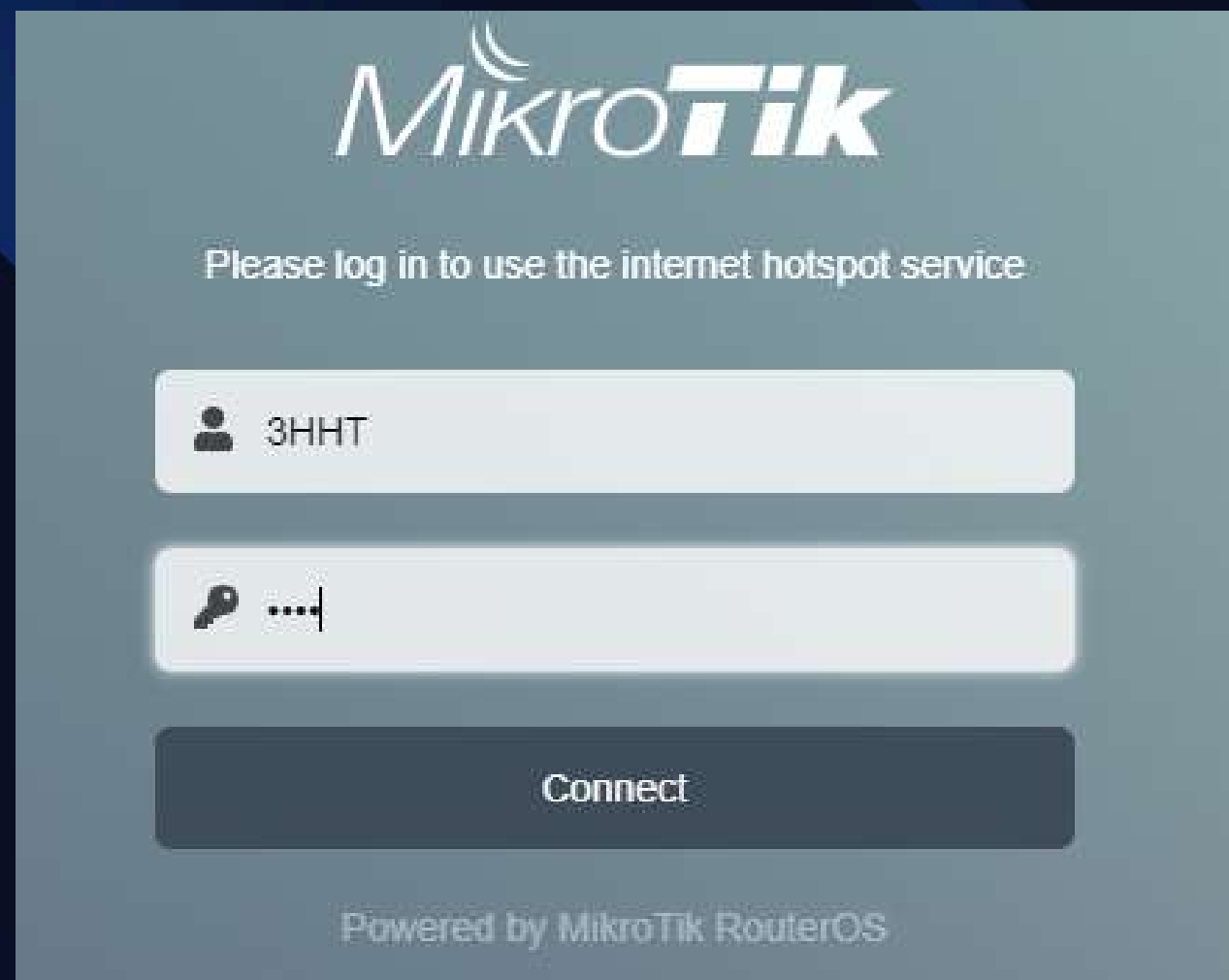
Comment

Validity : 1d | Price : Rp 5.000 | Selling Price : Rp 5.000 | Lock User : Disable

Kemudian akan muncul voucher seperti ini.

MIKHMON	
KELOMPOK 7 ID	
[1]	
Username	Password
3HHT	4673
1d 1d Rp 5.000	
Login: http://kelompok7.net	

Login menggunakan voucher tadi



The image shows the MikroTik login interface. At the top is the MikroTik logo. Below it, a message says "Please log in to use the internet hotspot service". There are two input fields: the first contains the username "3HHT" and the second contains a masked password "....". A "Connect" button is located below the password field. At the bottom, it says "Powered by MikroTik RouterOS".

MikroTik

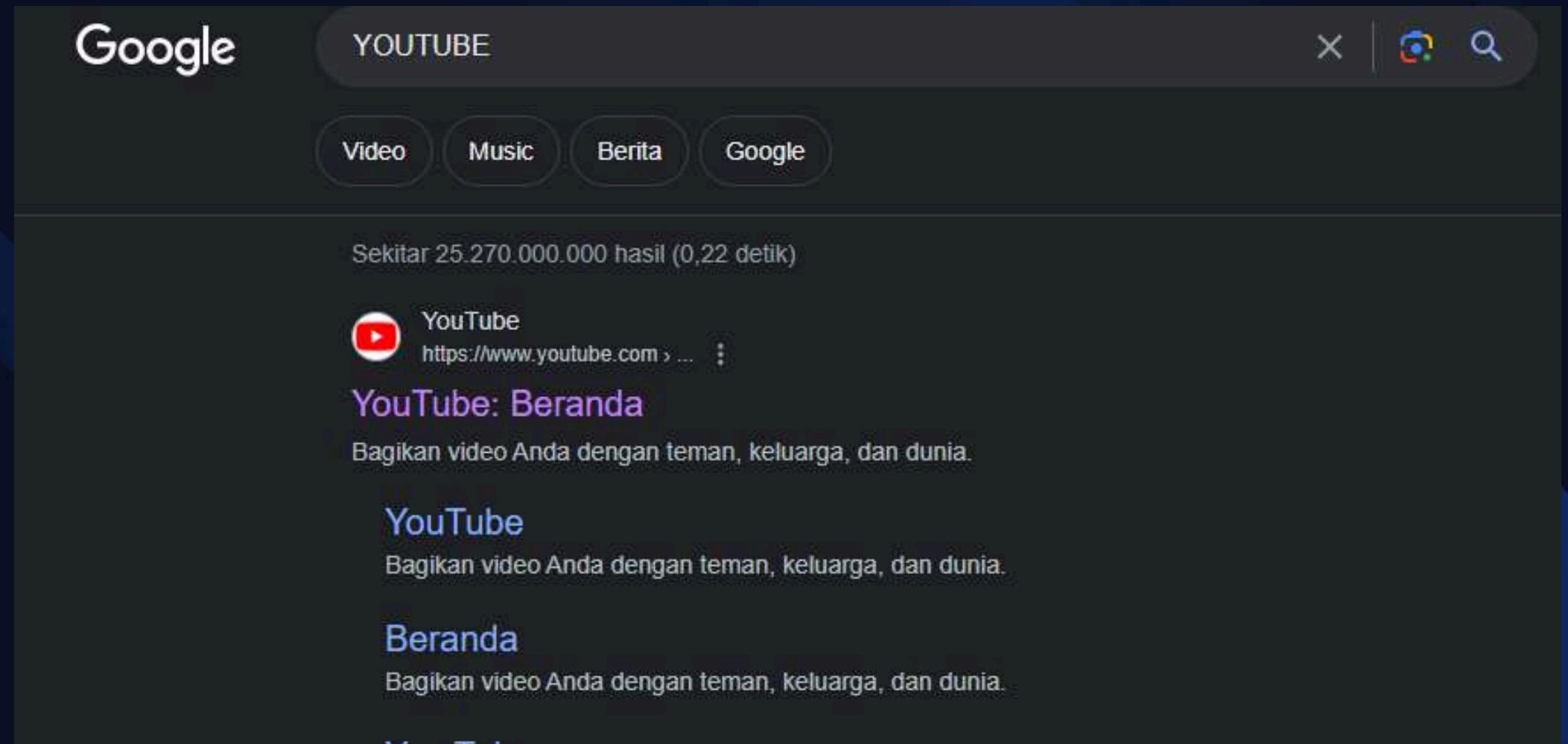
Please log in to use the internet hotspot service

3HHT

....

Connect

Powered by MikroTik RouterOS



internet berhasil

Kembali ke Mikrotik, sekarang atur waktu penggunaan di firewall NAT

NAT Rule <>

General Advanced Extra Action Statistics

Connection Limit

Limit

Dst. Limit

Nth

Time

Src. Address Type

Dst. Address Type

PSD

Hotspot

IP Fragment

Time: 07:00:00

- 16:00:00

Days: ☒ sat ☒ fri ☒ thu ☒ wed ☒ tue ☒ mon ☒ sun

FIREWALL LINUX.ORG

New Firewall Rule

General Advanced Extra Action ...

Src. Address List:

Dst. Address List:

Layer7 Protocol:

Content: ☐ linux.org

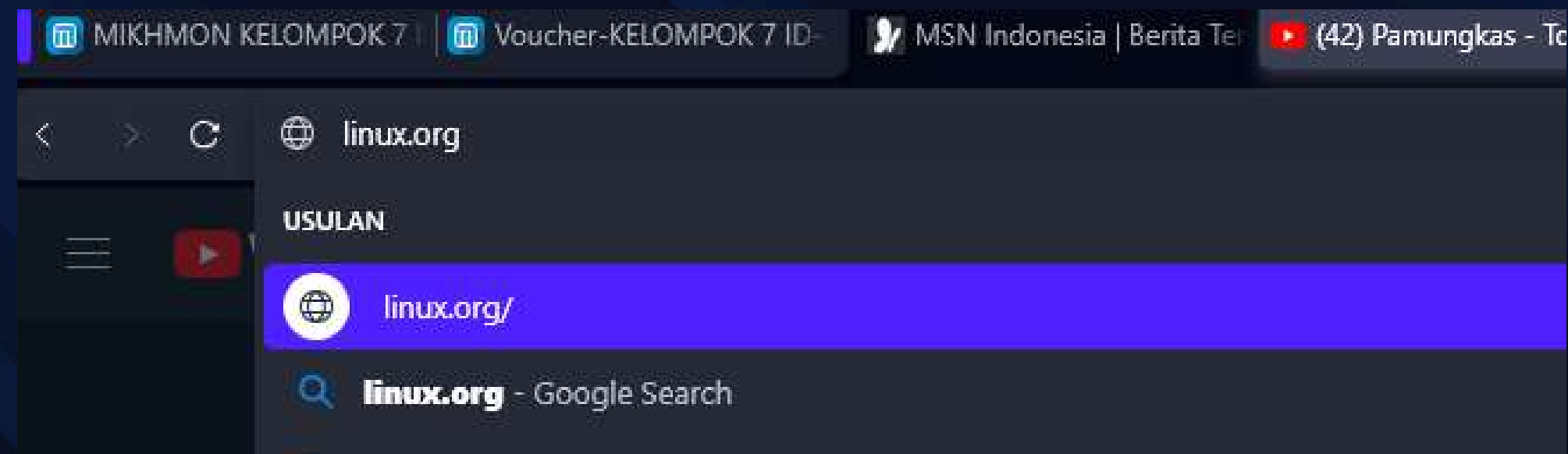
New Firewall Rule

Advanced Extra Action Statistics ...

Action: drop

☐ Log

Log Prefix:



FIREWALL .MP3 DAN .MKV

Firewall Rule <>

General Advanced Extra Action ...

Src. Address List: ▼

Dst. Address List: ▼

Layer7 Protocol: ▼

Content: ☐ .mp3 ▲

New Firewall Rule

General Advanced Extra Action ...

Src. Address List: ▼

Dst. Address List: ▼

Layer7 Protocol: ▼

Content: ☐ .mkv ▲

New Firewall Rule

Advanced Extra Action Statistics ...

Action: drop ▼

☐ Log

Log Prefix: ▼

BACKUP FILE

Backup

Name: Kelompok 7

Password:

Encryption: aes-sha256

☐ Don't Encrypt

Backup

Cancel

File List

File Cloud Backup

Backup Restore Upload...

Find

File Name	Type	Size	Creation Time
GLEN.backup	backup	11.9 KiB	Feb/23/2024 12:39:06
HANNI.backup	backup	16.3 KiB	Feb/23/2024 12:25:19
Juan Akay .backup	backup	11.2 KiB	Feb/13/2024 09:30:21
NickyKainde.backup	backup	11.2 KiB	Feb/13/2024 09:58:48
auto-before-reset.backup	backup	17.7 KiB	Jan/01/1970 08:00:08
glorianelw.backup	backup	13.1 KiB	Feb/23/2024 12:56:22
hizkiatumengkol.backup	backup	27.3 KiB	Apr/05/2023 12:30:30
hotspot	directory		Mar/31/2023 10:28:25
hotspot/alogin.html	.html file	1094 B	Mar/31/2023 10:28:25
hotspot/api.json	.json file	311 B	Mar/31/2023 10:28:25
hotspot/css	directory		Mar/31/2023 10:28:25
hotspot/css/style.css	.css file	4053 B	Mar/31/2023 10:28:25
hotspot/error.html	.html file	640 B	Mar/31/2023 10:28:25
hotspot/errors.txt	.txt file	3719 B	Mar/31/2023 10:28:25
hotspot/favicon.ico	.ico file	903 B	Mar/31/2023 10:28:25
hotspot/img	directory		Mar/31/2023 10:28:25
hotspot/img/password.svg	.svg file	644 B	Mar/31/2023 10:28:25

41 items

8.7 MiB of 16.0 MiB used

45% free

**TERIMA
KASIH**

